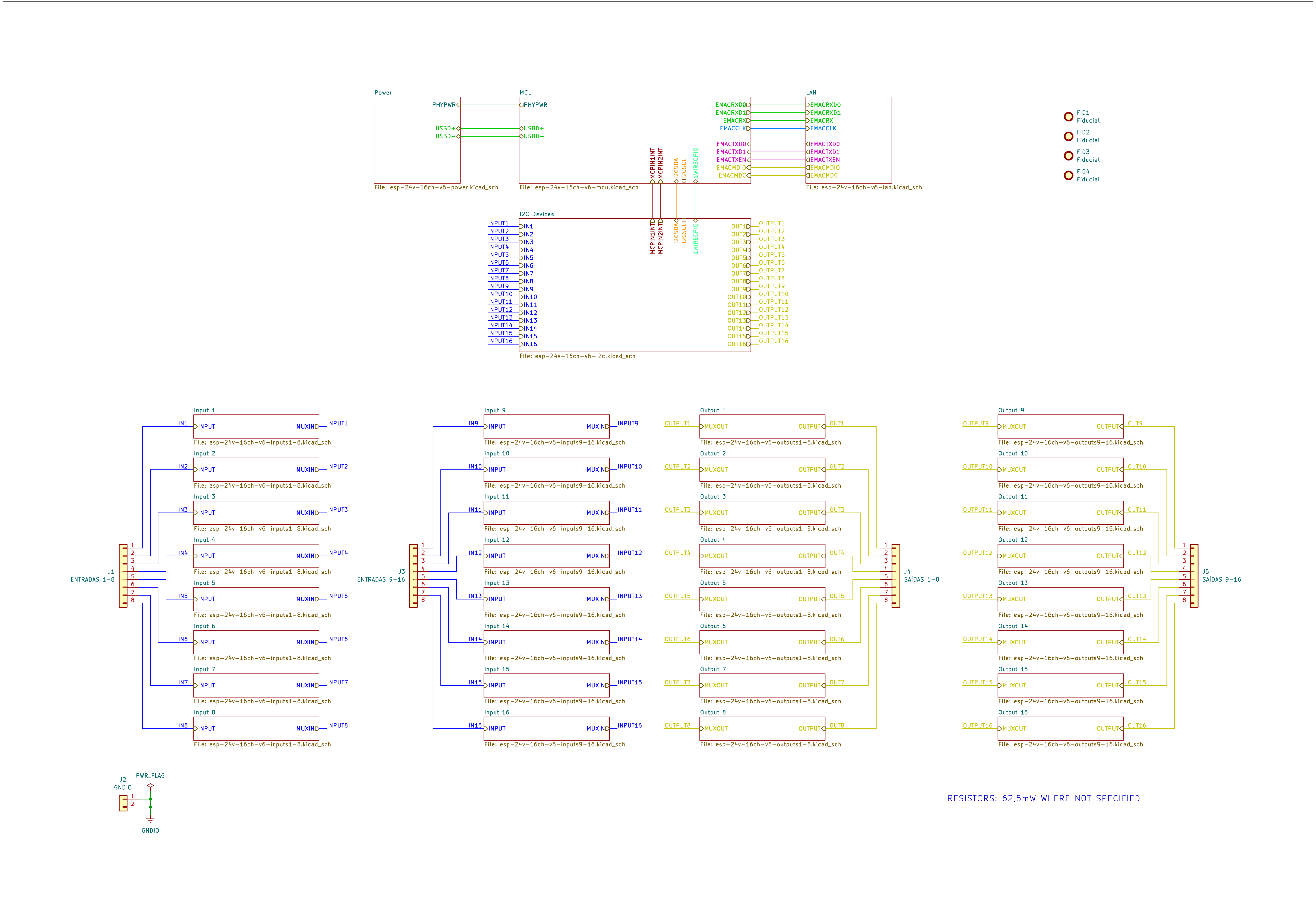
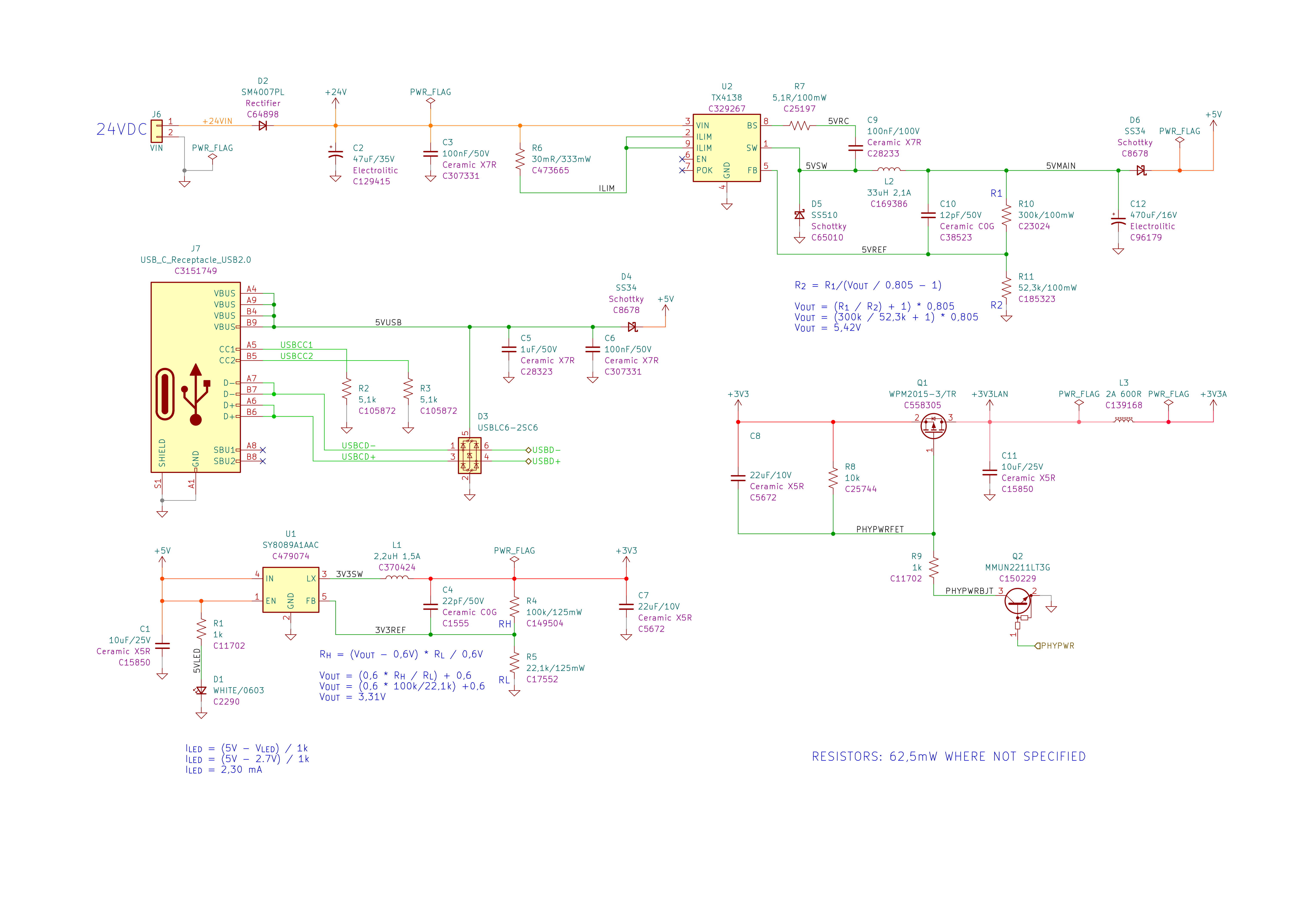
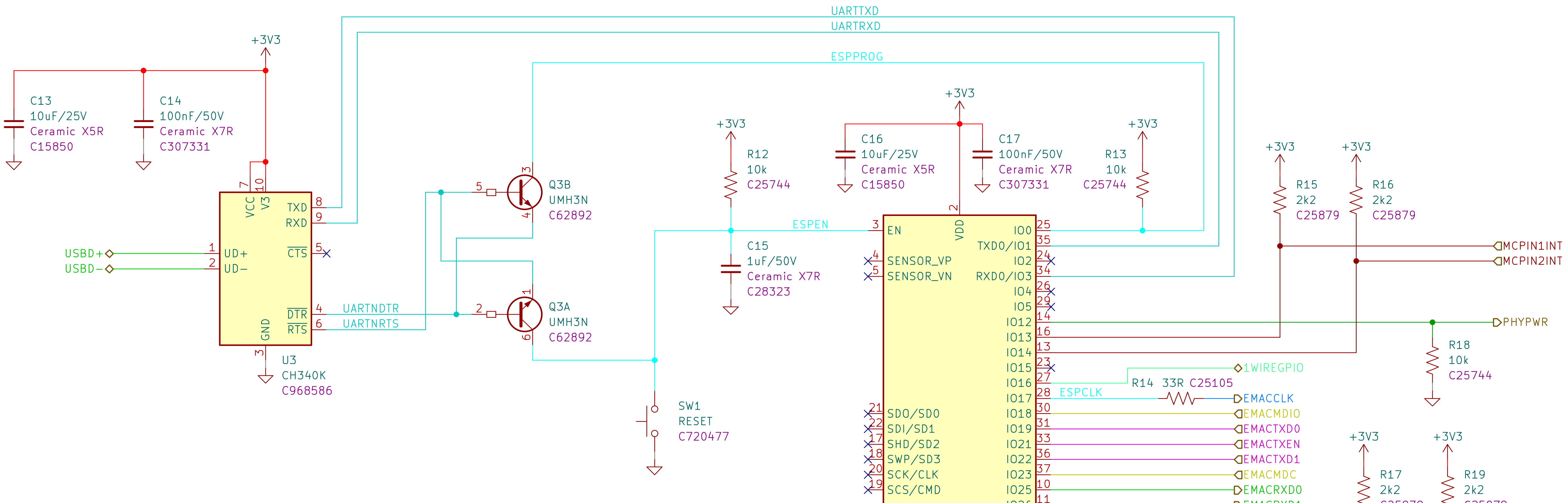
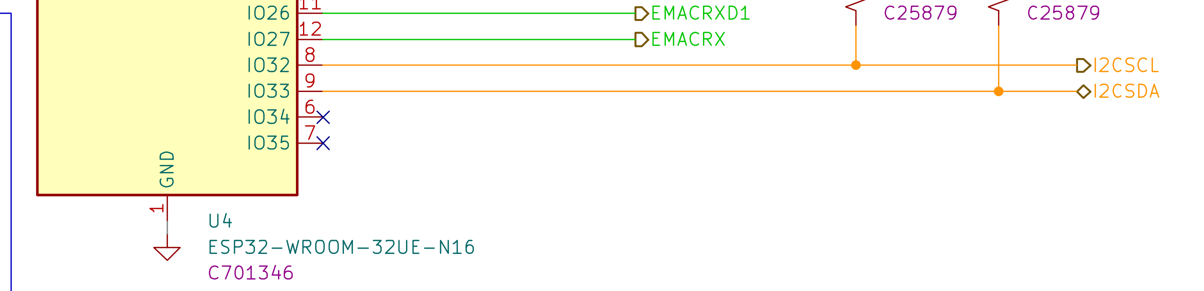


Power
PHYPWR
USBD+
USBD-
File: esp-24v-16ch-v6-power.kicad_sch
MCU
PHYPWR
USBD+
USBD-
EMACRXD0
EMACRXD1
EMACRX
EMACCLK
EMACTXD0
EMACTXD1
EMACTXEN
EMACMDIO
EMACMDC
MCPIN1INT
MCPIN2INT
I2CSDA
I2CSCL
1WIREGPIO
File: esp-24v-16ch-v6-mcu.kicad_sch
LAN
EMACRXD0
EMACRXD1
EMACRX
EMACCLK
EMACTXD0
EMACTXD1
EMACTXEN
EMACMDIO
EMACMDC
File: esp-24v-16ch-v6-lan.kicad_sch
FID1 Fiducial
FID2 Fiducial
FID3 Fiducial
FID4 Fiducial
I2C Devices
INPUT1 … INPUT16
IN1 … IN16
MCPIN1INT
MCPIN2INT
I2CSDA
I2CSCL
1WIREGPIO
OUT1 … OUT16
OUTPUT1 … OUTPUT16
File: esp-24v-16ch-v6-i2c.kicad_sch
Input 1 … Input 8
INPUT
MUXIN
File: esp-24v-16ch-v6-inputs1-8.kicad_sch
J1
ENTRADAS 1-8
Input 9 … Input 16
File: esp-24v-16ch-v6-inputs9-16.kicad_sch
J3
ENTRADAS 9-16
Output 1 … Output 8
MUXOUT
OUTPUT
File: esp-24v-16ch-v6-outputs1-8.kicad_sch
J4
SAÍDAS 1-8
Output 9 … Output 16
File: esp-24v-16ch-v6-outputs9-16.kicad_sch
J5
SAÍDAS 9-16
J2
GNDIO
PWR_FLAG
GNDIO
RESISTORS: 62,5mW WHERE NOT SPECIFIED


24VDC
J6
VIN
D2
SM4007PL
Rectifier
C64898
+24VIN
+24V
PWR_FLAG
C2
47uF/35V
Electrolitic
C129415
C3
100nF/50V
Ceramic X7R
C307331
R6
30mR/333mW
C473665
ILIM
U2
TX4138
C329267
VIN
ILIM
ILIM
EN
POK
GND
BS
SW
FB
R7
5,1R/100mW
C25197
5VRC
C9
100nF/100V
Ceramic X7R
C28233
5VSW
L2
33uH 2,1A
C169386
D5
SS510
Schottky
C65010
C10
12pF/50V
Ceramic C0G
C38523
5VREF
R1
R10
300k/100mW
C23024
R11
52,3k/100mW
C185323
R2
5VMAIN
D6
SS34
Schottky
C8678
+5V
C12
470uF/16V
Electrolitic
C96179
R2 = R1/(VOUT / 0,805 - 1)
VOUT = (R1 / R2) + 1) * 0,805
VOUT = (300k / 52,3k + 1) * 0,805
VOUT = 5,42V
J7
USB_C_Receptacle_USB2.0
C3151749
VBUS
CC1
CC2
D-
D+
SBU1
SBU2
SHIELD
GND
5VUSB
USBCC1
USBCC2
D4
SS34
Schottky
C8678
C5
1uF/50V
Ceramic X7R
C28323
C6
100nF/50V
Ceramic X7R
C307331
R2
5,1k
C105872
R3
5,1k
C105872
D3
USBLC6-2SC6
USBCD-
USBCD+
USBD-
USBD+
U1
SY8089A1AAC
C479074
IN
EN
GND
LX
FB
3V3SW
L1
2,2uH 1,5A
C370424
3V3REF
C1
10uF/25V
Ceramic X5R
C15850
R1
1k
C11702
5VLED
D1
WHITE/0603
C2290
C4
22pF/50V
Ceramic C0G
C1555
R4
100k/125mW
C149504
RH
R5
22,1k/125mW
C17552
RL
+3V3
C7
22uF/10V
Ceramic X5R
C5672
RH = (VOUT - 0,6V) * RL / 0,6V
VOUT = (0,6 * RH / RL) + 0,6
VOUT = (0,6 * 100k/22,1k) +0,6
VOUT = 3,31V
ILED = (5V - VLED) / 1k
ILED = (5V - 2.7V) / 1k
ILED = 2,30 mA
Q1
WPM2015-3/TR
C558305
+3V3LAN
L3
2A 600R
C139168
+3V3A
C8
22uF/10V
Ceramic X5R
C5672
R8
10k
C25744
C11
10uF/25V
Ceramic X5R
C15850
PHYPWRFET
R9
1k
C11702
Q2
MMUN2211LT3G
C150229
PHYPWRBJT
PHYPWR
RESISTORS: 62,5mW WHERE NOT SPECIFIED





TEMPLATE FOR GPIO ASSIGNMENT

{"NAME":"ESP-24V-16CH-V6","GPIO":[0,3200,0,3232,0,0,0,0,5536,10240,10241,0,0,
0,5600,0,0,0,0,5568,0,0,0,0,0,0,0,0,608,640,192,193,0,0,0,0],"FLAG":0,"BASE":1}

See https://tasmota.github.io/docs/Templates/

TEMPLATE FOR MCP23017 I/O ASSIGNMENT

{"NAME":"MCP23017 expander","BASE":0,"GPIO":[192,193,194,195,196,197,198,0,199,200,201,
202,203,204,205,0,224,225,226,227,228,229,230,231,232,233,234,235,236,237,238,239],
"IOCON":0x5C}

Save as mcp23x.dat and load into tasmota's filesystem.
See https://tasmota.github.io/docs/MCP230xx/

TASMOTA CONFIGS
The configs below must be applied via terminal/console and adjusted according to the
desired hostname, device name, network addresses and behavior of switches.

Configure switches:
> backlog switchmode1 5; switchmode2 5; switchmode3 5; switchmode4 5; switchmode5 5;
switchmode6 5; switchmode7 5; switchmode8 5; switchmode9 5; switchmode10 5;
switchmode11 5; switchmode12 5; switchmode13 5; switchmode14 5; switchmode15 5;
switchmode16 5

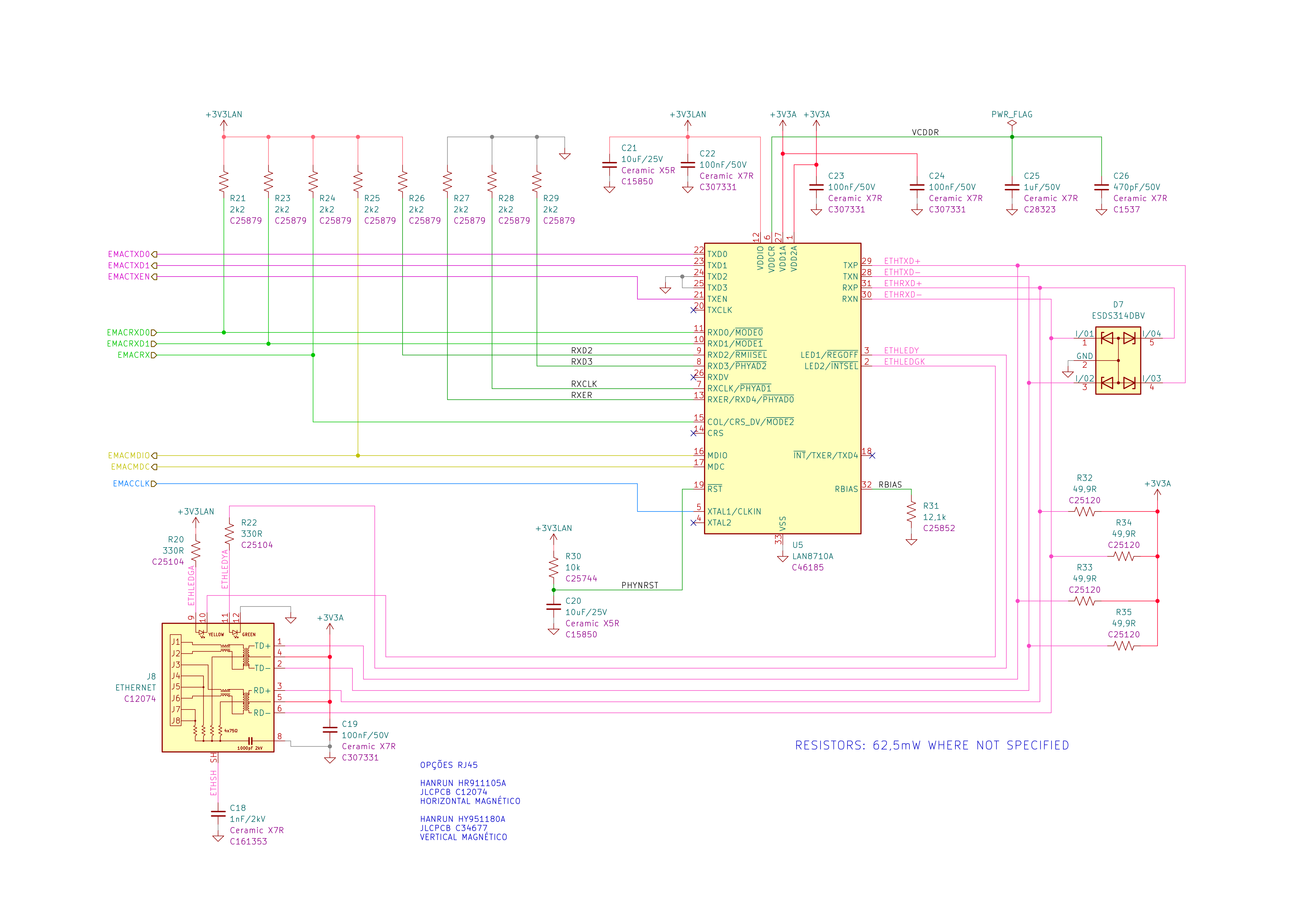
Configure device name:
> devicename fararmio066022

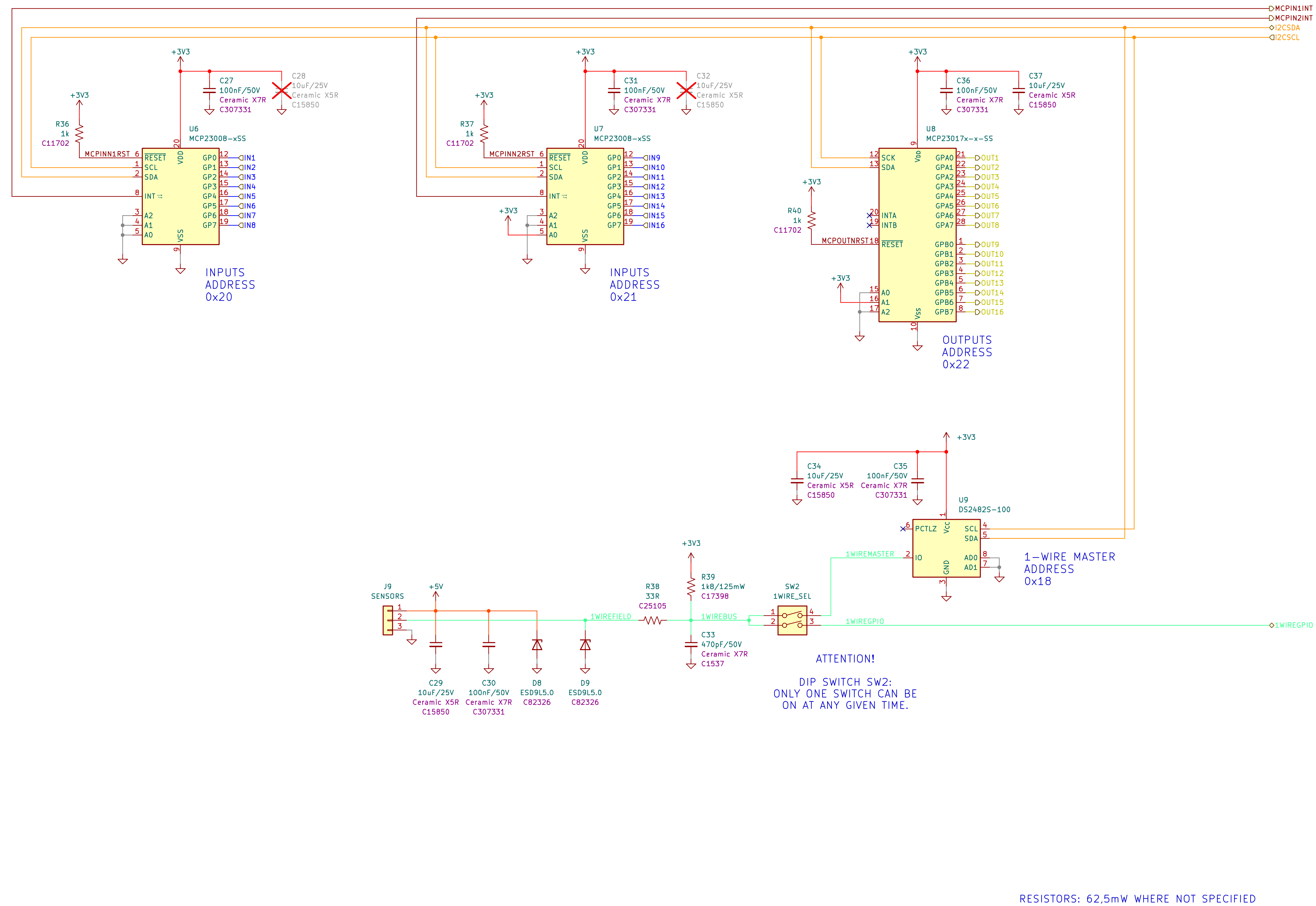
Configure network:
> backlog ethipaddress 10.112.66.22; ethsubnetmask 255.255.255.0; ethgateway 10.112.66.254;
wifi 0; hostname fararmio066022; topic fararmio066022; devicename fararmio066022

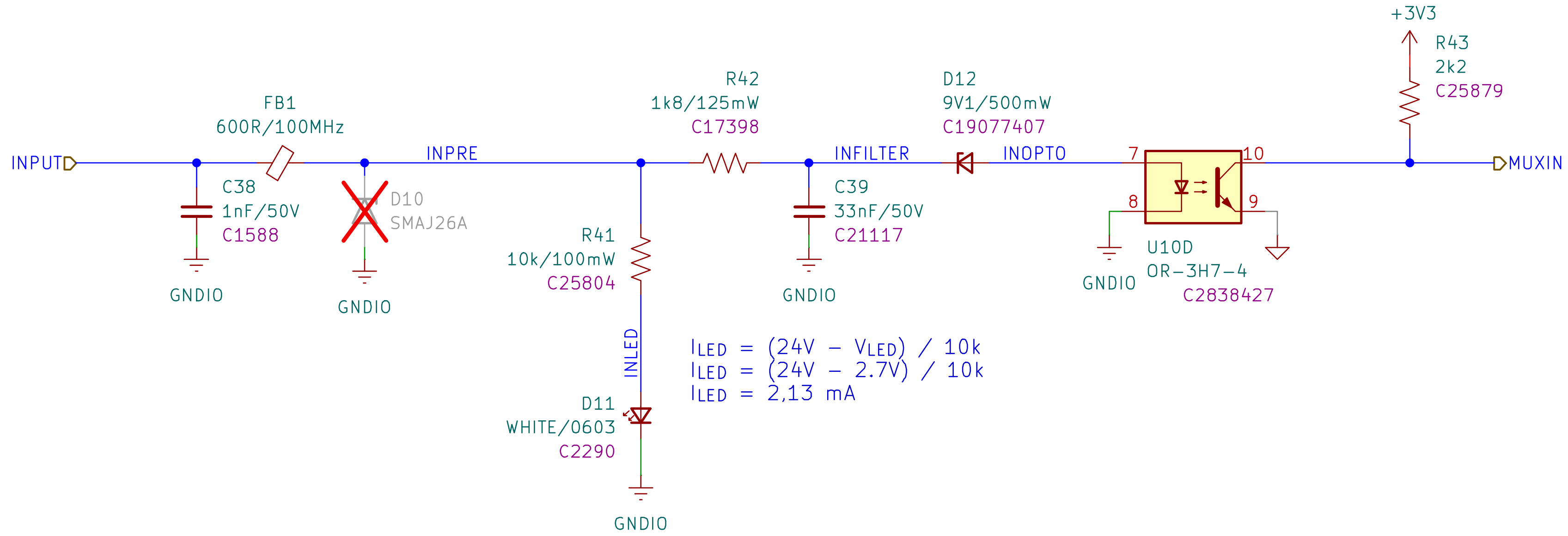
ESP32 GPIO ASSIGNMENT

PIN	PORT	FUNCTION	COMPONENT
25	GPIO0	AUTO FLASH MODE	0
35	GPIO1	SERIAL TX	3200
24	GPIO2	–	0
34	GPIO3	SERIAL RX	3232
26	GPIO4	–	0
29	GPIO5	–	0
17	GPIO9	–	0
18	GPIO10	–	0
14	GPIO12	ETH POWER	5536
16	GPIO13	MCP23017 INT1	10240
13	GPIO14	MCP23017 INT2	10241
23	GPIO15	–	0
27	GPIO16	–	0
28	GPIO17	ETH CLK	0
30	GPIO18	ETH MDIO	5600
31	GPIO19	ETH TXD0	0
–	GPIO20	–	0
33	GPIO21	ETH TXEN	0
36	GPIO22	ETH TXD1	0
37	GPIO23	ETH MDC	5568
–	GPIO24	–	0
10	GPIO25	ETH RXD0	0
11	GPIO26	ETH RXD1	0
12	GPIO27	ETH RX	0
20	GPIO6	–	0
21	GPIO7	–	0
22	GPIO8	–	0
19	GPIO11	–	0
8	GPIO32	I2C SCL	608
9	GPIO33	I2C SDA	640
6	GPIO34	–	0
7	GPIO35	–	0
4	GPIO36	–	0
–	GPIO37	–	0
–	GPIO38	–	0
5	GPIO39	–	0

RESISTORS: 62,5mW WHERE NOT SPECIFIED

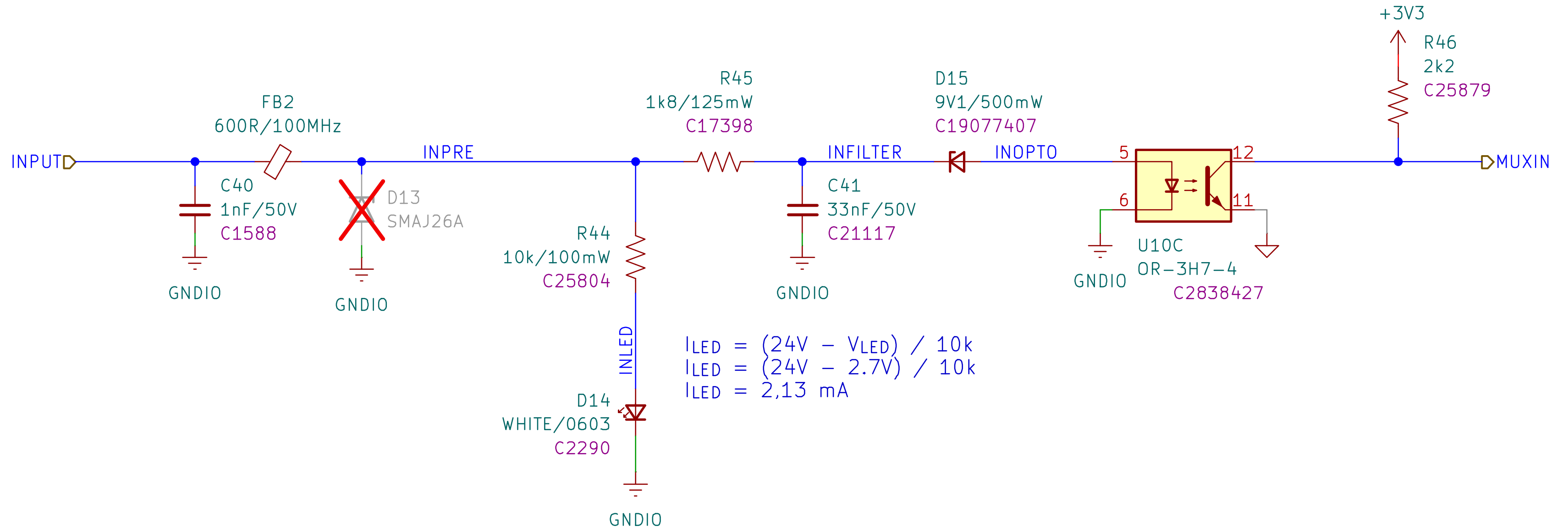

+3V3LAN
R21 2k2 C25879
R23 2k2 C25879
R24 2k2 C25879
R25 2k2 C25879
R26 2k2 C25879
R27 2k2 C25879
R28 2k2 C25879
R29 2k2 C25879
C21 10uF/25V Ceramic X5R C15850
C22 100nF/50V Ceramic X7R C307331
+3V3A
VCDDR
PWR_FLAG
C23 100nF/50V Ceramic X7R C307331
C24 100nF/50V Ceramic X7R C307331
C25 1uF/50V Ceramic X7R C28323
C26 470pF/50V Ceramic X7R C1537
EMACTXD0
EMACTXD1
EMACTXEN
EMACRXD0
EMACRXD1
EMACRX
EMACMDIO
EMACMDC
EMACCLK
TXD0
TXD1
TXD2
TXD3
TXEN
TXCLK
RXD0/MODE0
RXD1/MODE1
RXD2/RMIISEL
RXD3/PHYAD2
RXDV
RXCLK/PHYAD1
RXER/RXD4/PHYAD0
COL/CRS_DV/MODE2
CRS
MDIO
MDC
RST
XTAL1/CLKIN
XTAL2
VDDIO
VDDCR
VDD1A
VDD2A
VSS
TXP
TXN
RXP
RXN
LED1/REGOFF
LED2/INTSEL
INT/TXER/TXD4
RBIAS
RXD2
RXD3
RXCLK
RXER
ETHTXD+
ETHTXD−
ETHRXD+
ETHRXD−
ETHLEDY
ETHLEDGK
U5 LAN8710A C46185
D7 ESDS314DBV
I/01
I/02
I/03
I/04
GND
R31 12,1k C25852
R32 49,9R C25120
R33 49,9R C25120
R34 49,9R C25120
R35 49,9R C25120
R30 10k C25744
PHYNRST
C20 10uF/25V Ceramic X5R C15850
R20 330R C25104
R22 330R C25104
ETHLEDGA
ETHLEDYA
YELLOW
GREEN
TD+
TD−
RD+
RD−
4x75Ω
1000pF 2kV
J8 ETHERNET C12074
SH
ETHSH
C18 1nF/2kV Ceramic X7R C161353
C19 100nF/50V Ceramic X7R C307331
RESISTORS: 62,5mW WHERE NOT SPECIFIED
OPÇÕES RJ45
HANRUN HR911105A
JLCPCB C12074
HORIZONTAL MAGNÉTICO
HANRUN HY951180A
JLCPCB C34677
VERTICAL MAGNÉTICO


MCPIN1INT
MCPIN2INT
I2CSDA
I2CSCL
+3V3
R36
1k
C11702
MCPINN1RST
U6
MCP23008-xSS
RESET
SCL
SDA
INT
A2
A1
A0
VDD
VSS
GP0
GP1
GP2
GP3
GP4
GP5
GP6
GP7
IN1
IN2
IN3
IN4
IN5
IN6
IN7
IN8
C27
100nF/50V
Ceramic X7R
C307331
C28
10uF/25V
Ceramic X5R
C15850
INPUTS
ADDRESS
0x20
R37
1k
C11702
MCPINN2RST
U7
MCP23008-xSS
IN9
IN10
IN11
IN12
IN13
IN14
IN15
IN16
C31
100nF/50V
Ceramic X7R
C307331
C32
10uF/25V
Ceramic X5R
C15850
INPUTS
ADDRESS
0x21
U8
MCP23017x-x-SS
SCK
SDA
INTA
INTB
RESET
A0
A1
A2
Vdd
Vss
GPA0
GPA1
GPA2
GPA3
GPA4
GPA5
GPA6
GPA7
GPB0
GPB1
GPB2
GPB3
GPB4
GPB5
GPB6
GPB7
OUT1
OUT2
OUT3
OUT4
OUT5
OUT6
OUT7
OUT8
OUT9
OUT10
OUT11
OUT12
OUT13
OUT14
OUT15
OUT16
C36
100nF/50V
Ceramic X7R
C307331
C37
10uF/25V
Ceramic X5R
C15850
R40
1k
C11702
MCPOUTNRST
OUTPUTS
ADDRESS
0x22
C34
10uF/25V
Ceramic X5R
C15850
C35
100nF/50V
Ceramic X7R
C307331
U9
DS2482S-100
PCTLZ
IO
Vcc
GND
SCL
SDA
AD0
AD1
1WIREMASTER
1-WIRE MASTER
ADDRESS
0x18
J9
SENSORS
+5V
1WIREFIELD
R38
33R
C25105
R39
1k8/125mW
C17398
1WIREBUS
C33
470pF/50V
Ceramic X7R
C1537
SW2
1WIRE_SEL
1WIREGPIO
C29
10uF/25V
Ceramic X5R
C15850
C30
100nF/50V
Ceramic X7R
C307331
D8
ESD9L5.0
C82326
D9
ESD9L5.0
C82326
ATTENTION!
DIP SWITCH SW2:
ONLY ONE SWITCH CAN BE
ON AT ANY GIVEN TIME.
RESISTORS: 62,5mW WHERE NOT SPECIFIED



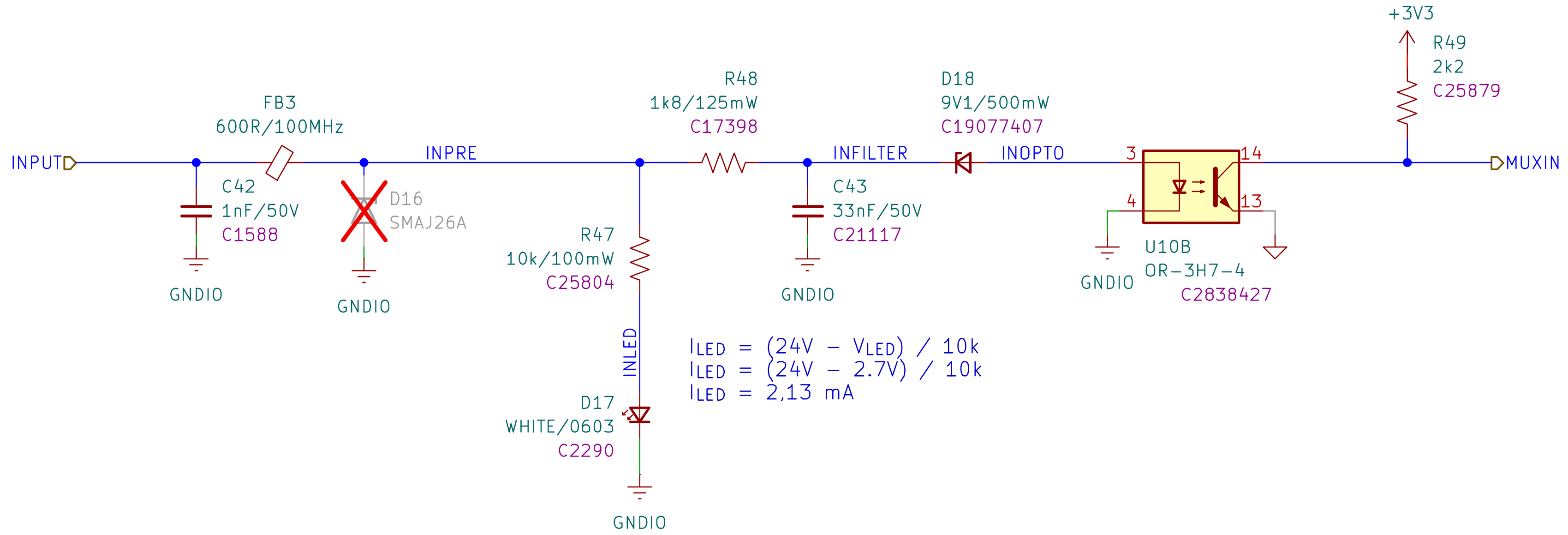
I_{LED} = (24V − V_{LED}) / 10k
I_{LED} = (24V − 2.7V) / 10k
I_{LED} = 2,13 mA

RESISTORS: 62,5mW WHERE NOT SPECIFIED

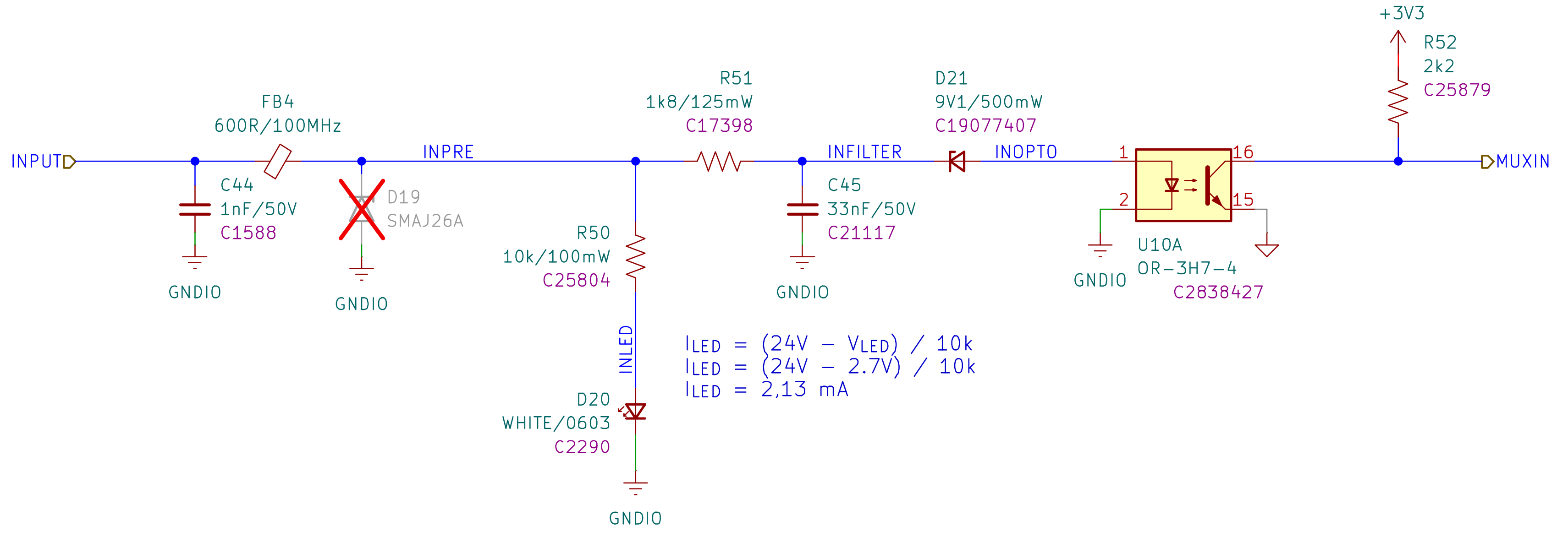


$I_{LED} = (24V - V_{LED}) / 10k$
$I_{LED} = (24V - 2.7V) / 10k$
$I_{LED} = 2,13$ mA

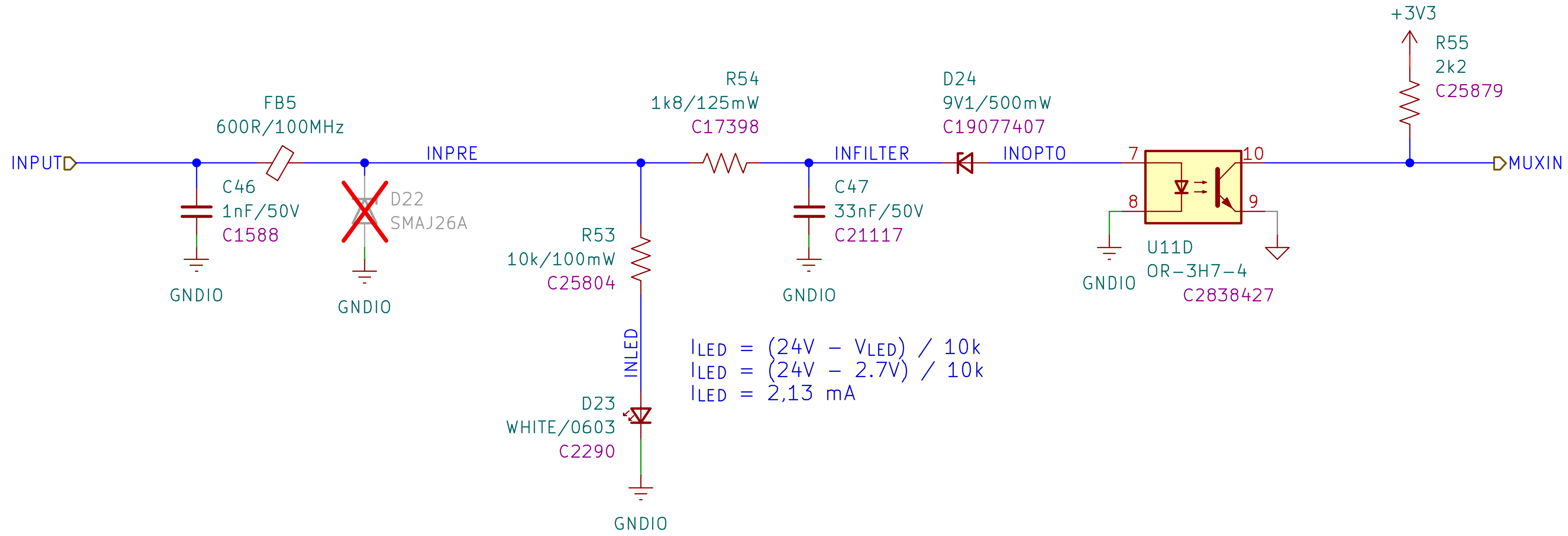
RESISTORS: 62,5mW WHERE NOT SPECIFIED



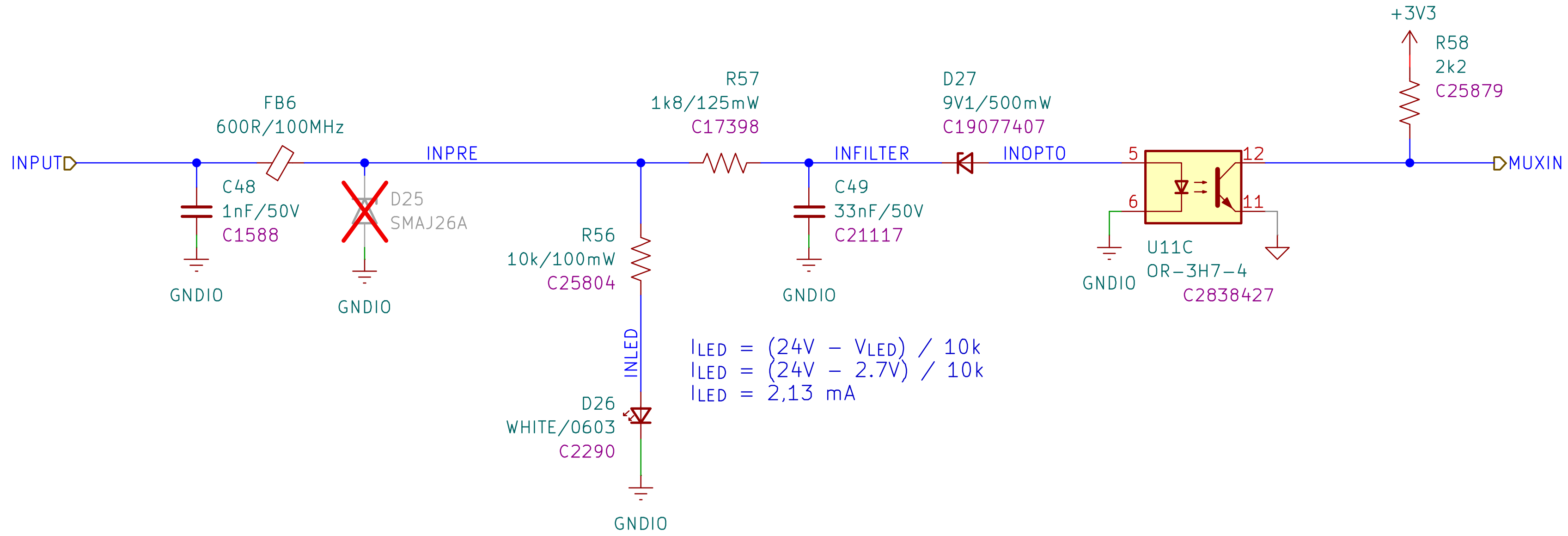
RESISTORS: 62,5mW WHERE NOT SPECIFIED



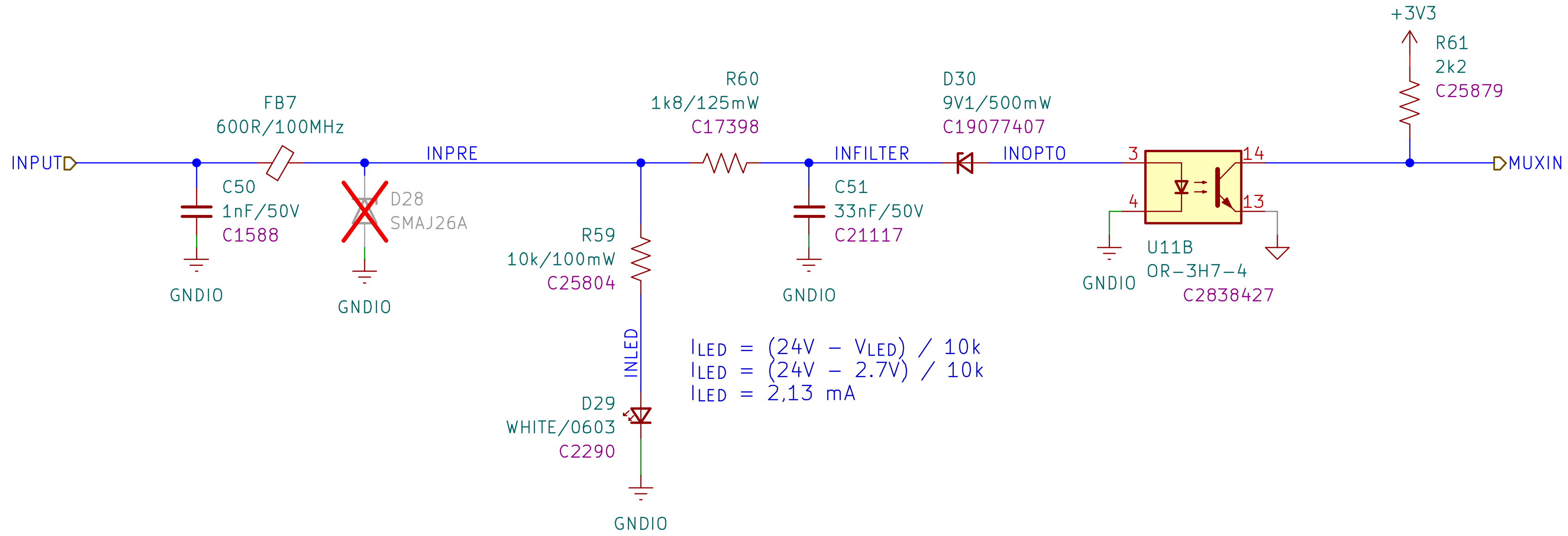
RESISTORS: 62,5mW WHERE NOT SPECIFIED



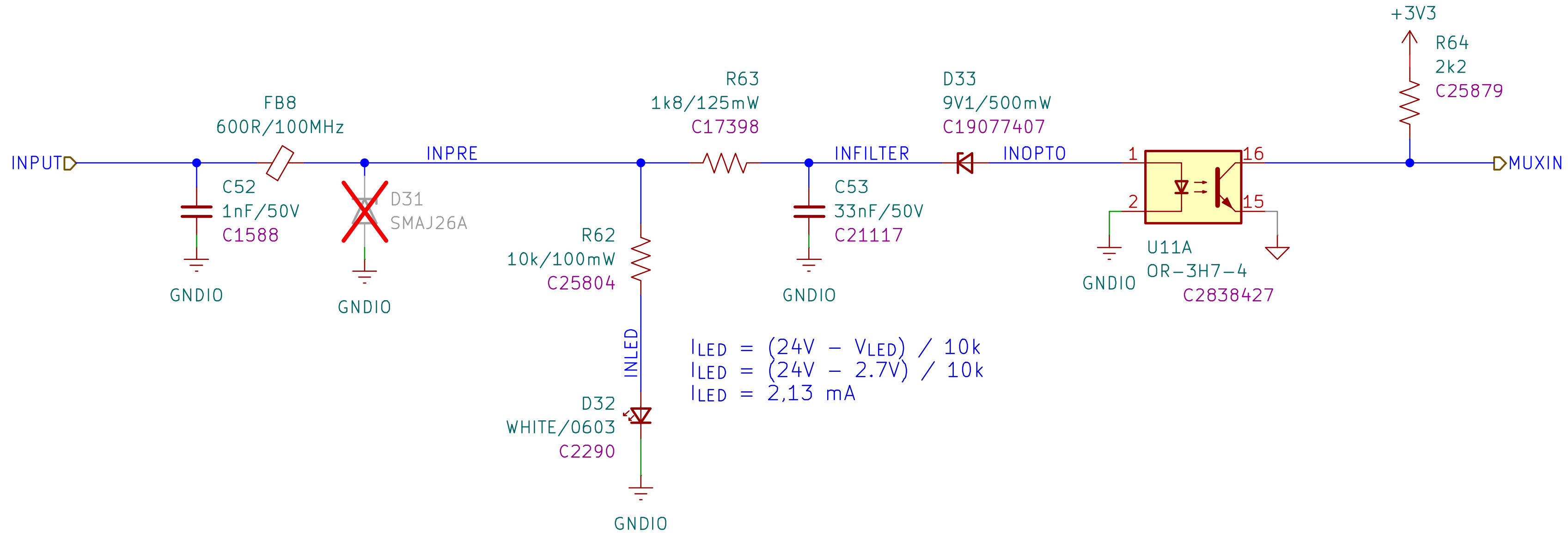
RESISTORS: 62,5mW WHERE NOT SPECIFIED



RESISTORS: 62,5mW WHERE NOT SPECIFIED

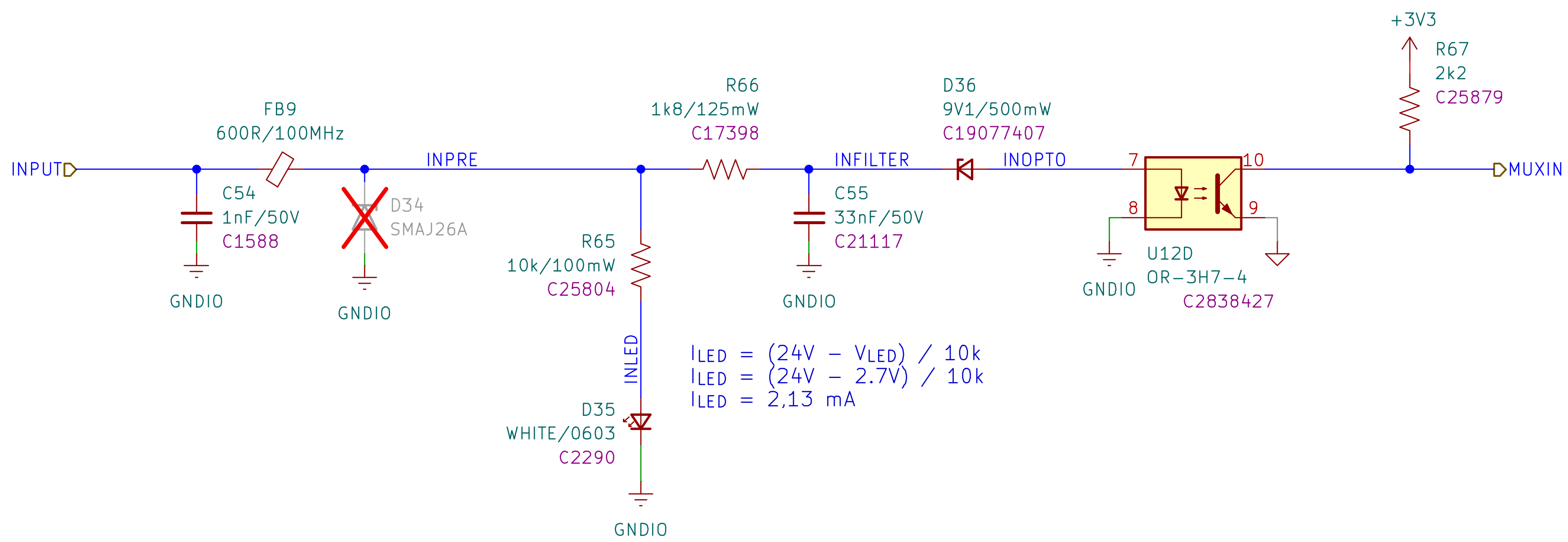


RESISTORS: 62,5mW WHERE NOT SPECIFIED

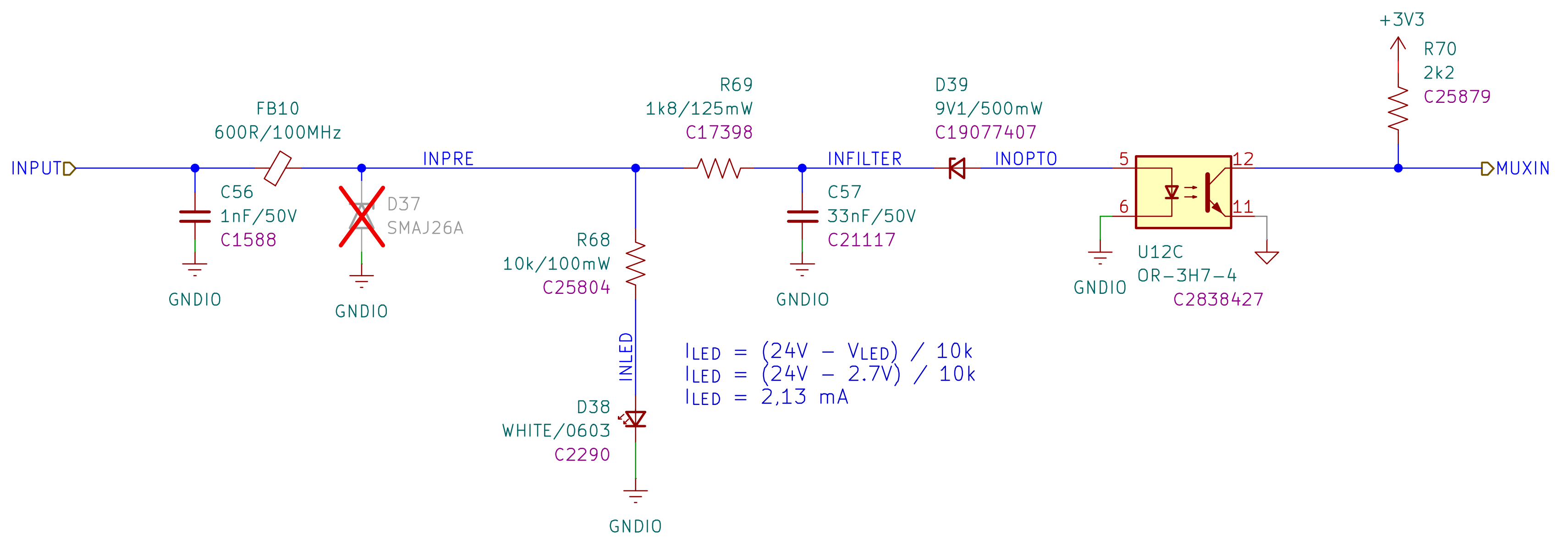


I_{LED} = (24V − V_{LED}) / 10k
I_{LED} = (24V − 2.7V) / 10k
I_{LED} = 2,13 mA

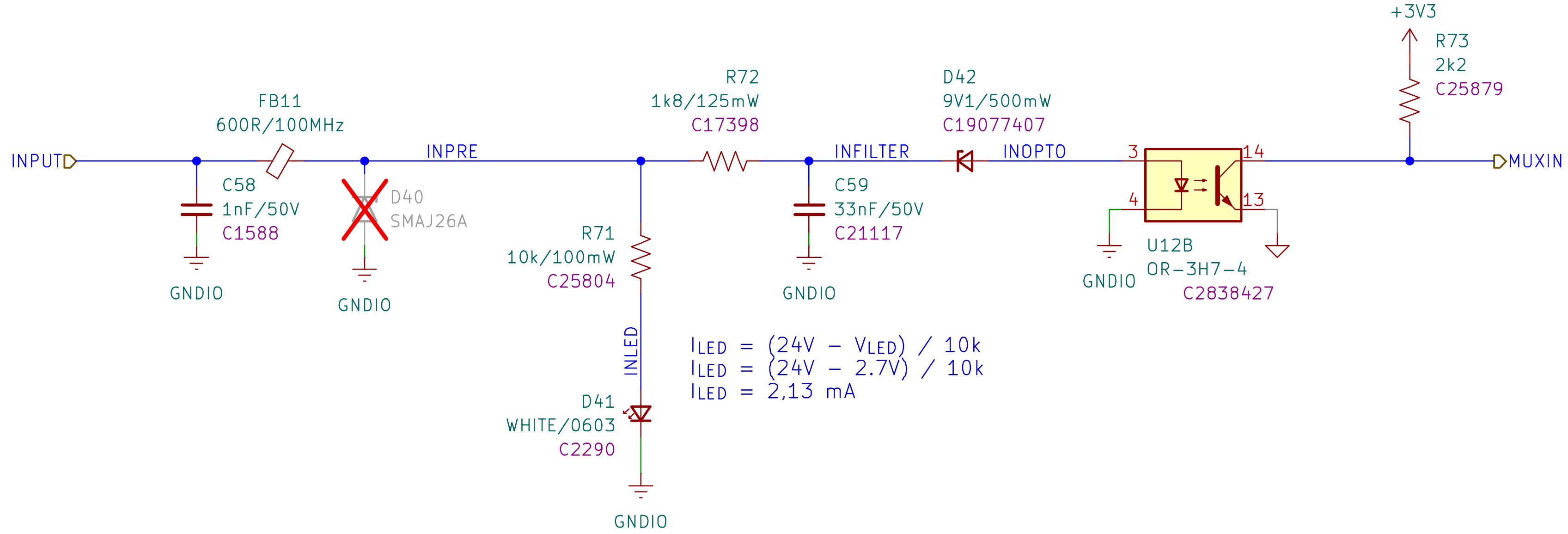
RESISTORS: 62,5mW WHERE NOT SPECIFIED



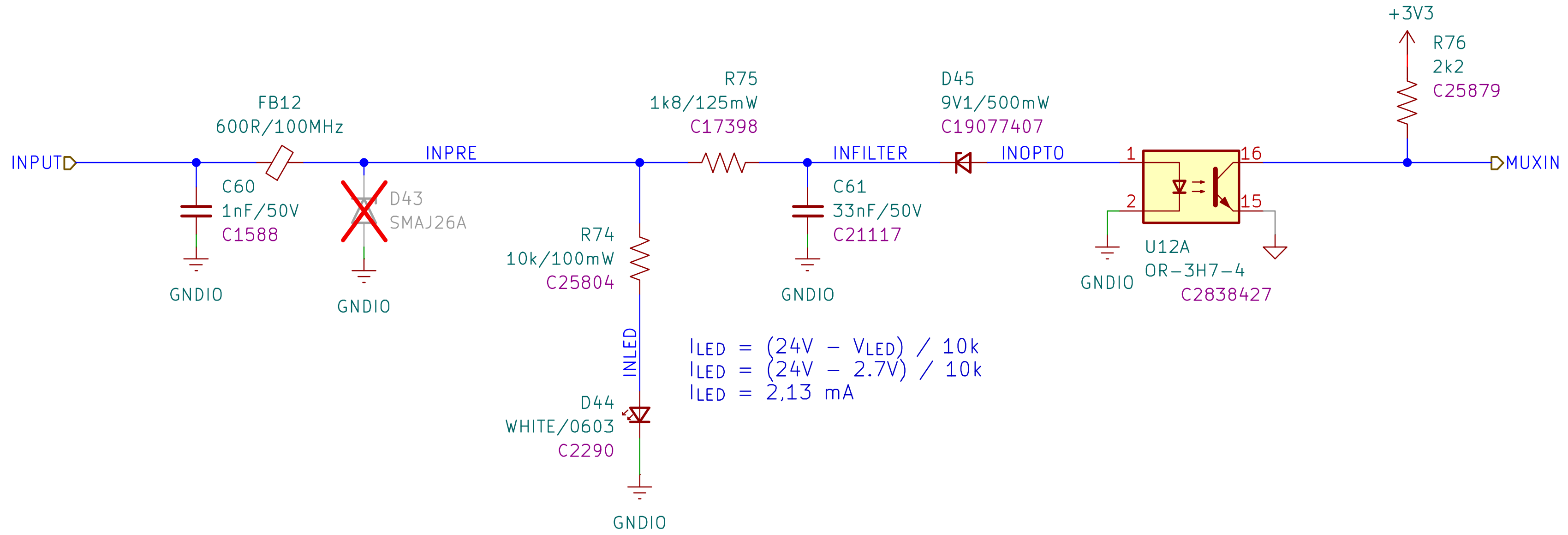
RESISTORS: 62,5mW WHERE NOT SPECIFIED



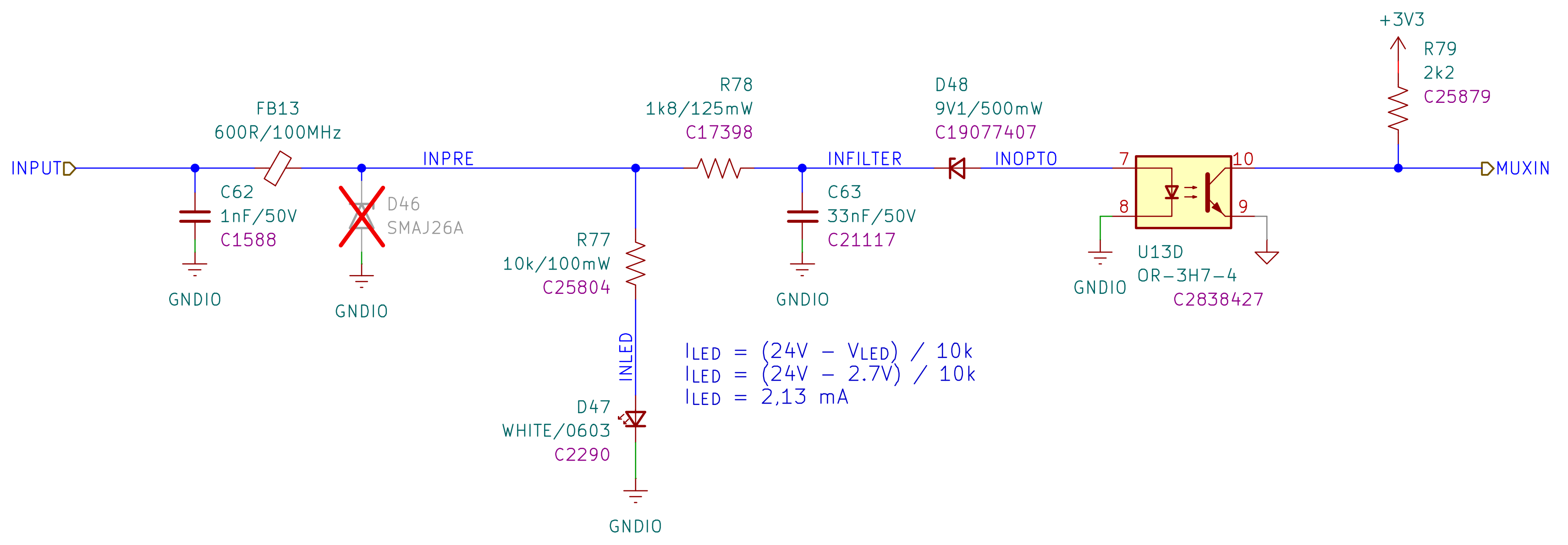
RESISTORS: 62,5mW WHERE NOT SPECIFIED



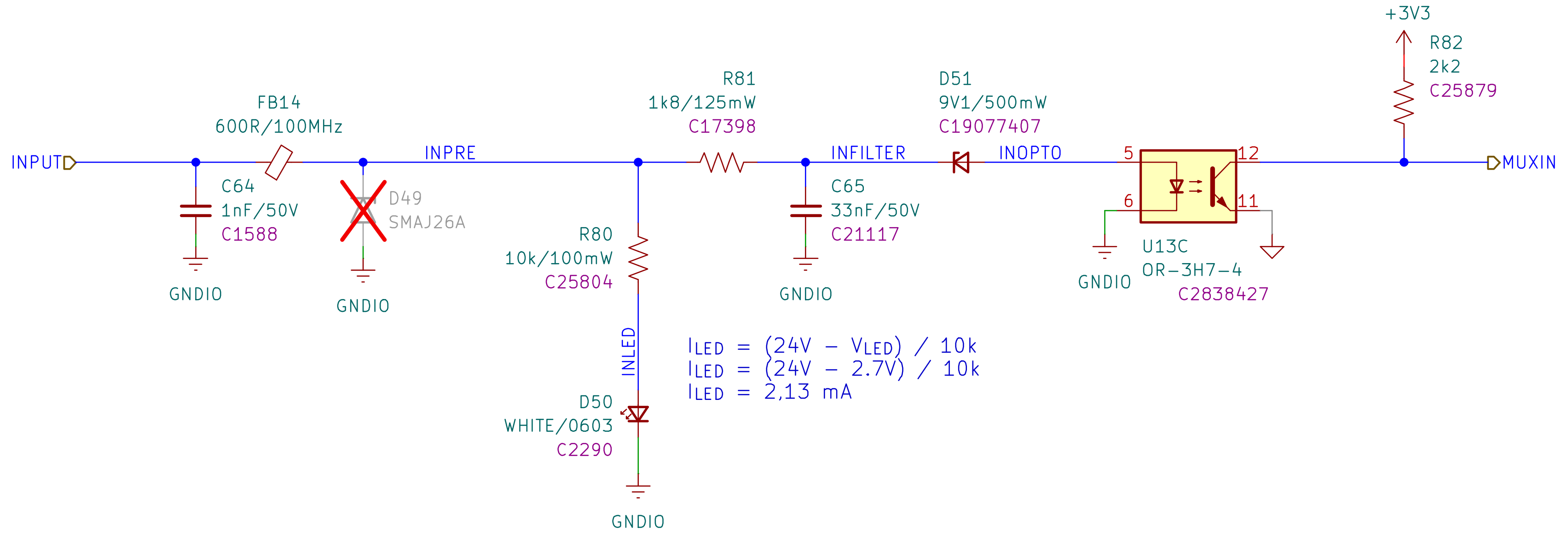
RESISTORS: 62,5mW WHERE NOT SPECIFIED



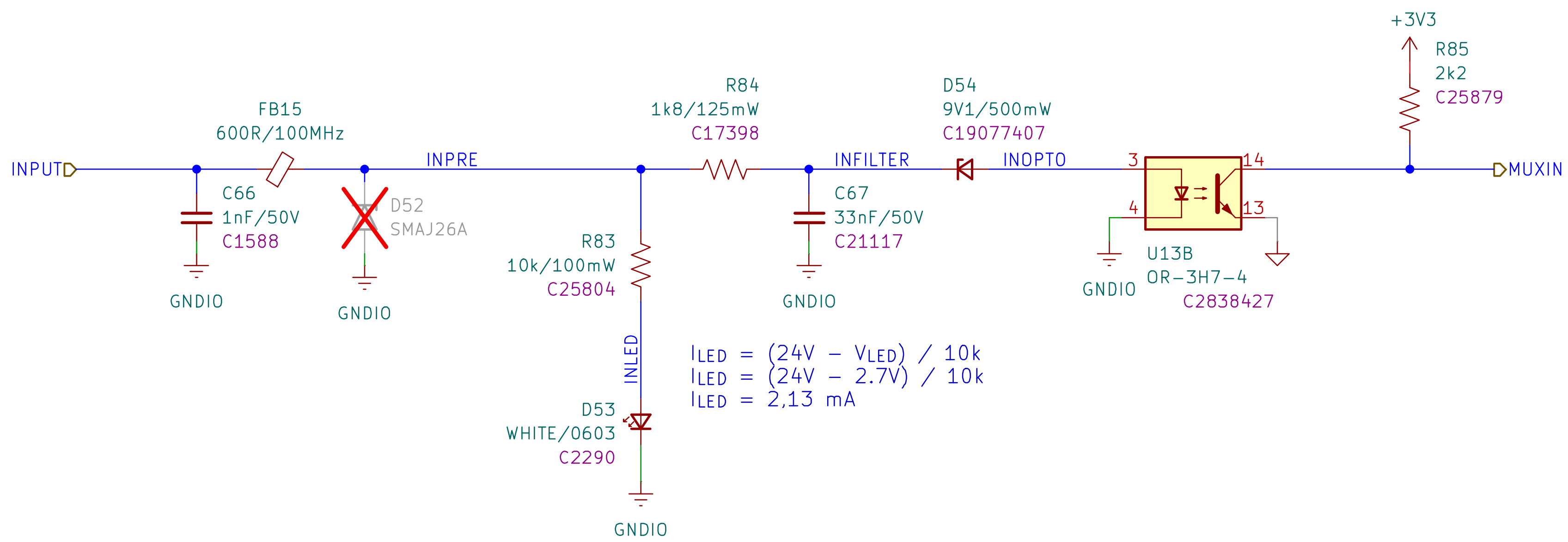
RESISTORS: 62,5mW WHERE NOT SPECIFIED



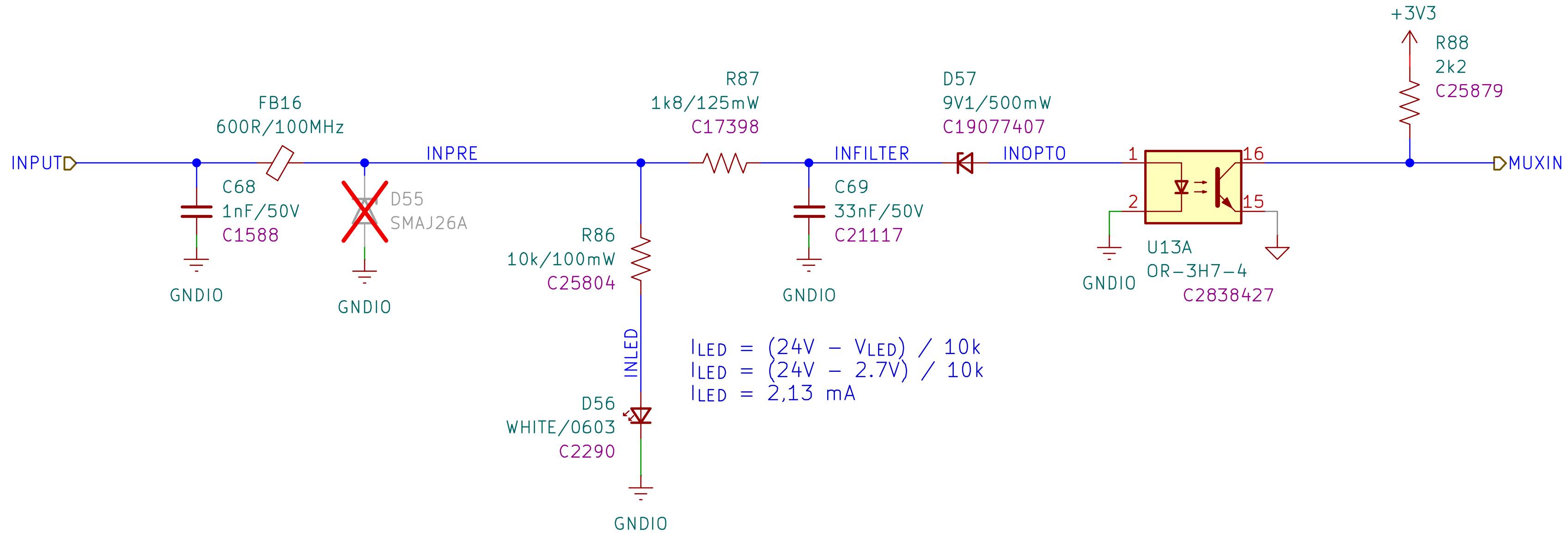
RESISTORS: 62,5mW WHERE NOT SPECIFIED



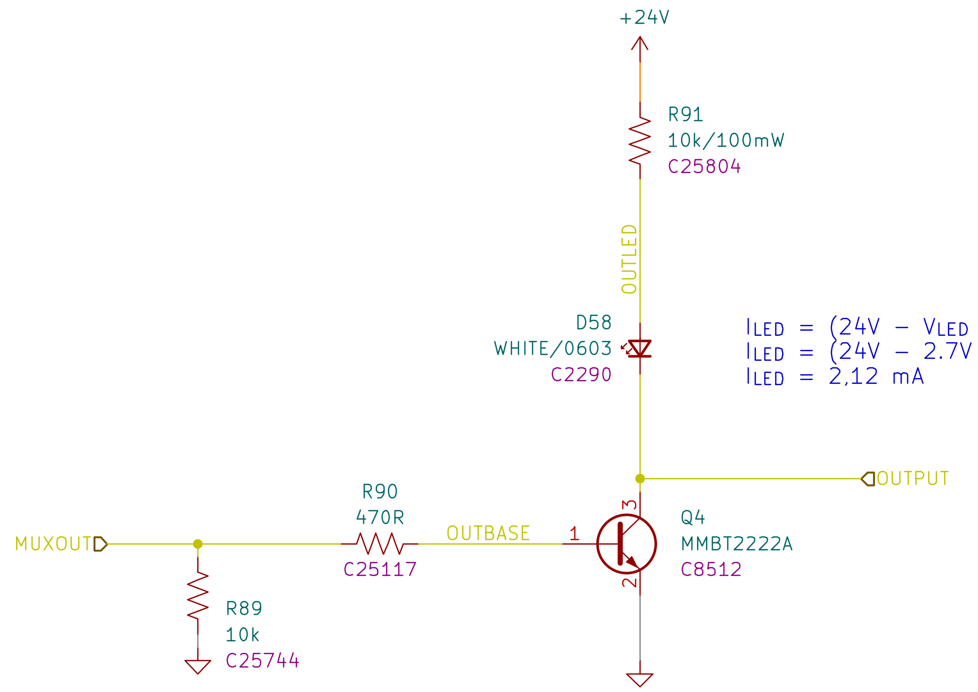
RESISTORS: 62,5mW WHERE NOT SPECIFIED



RESISTORS: 62,5mW WHERE NOT SPECIFIED



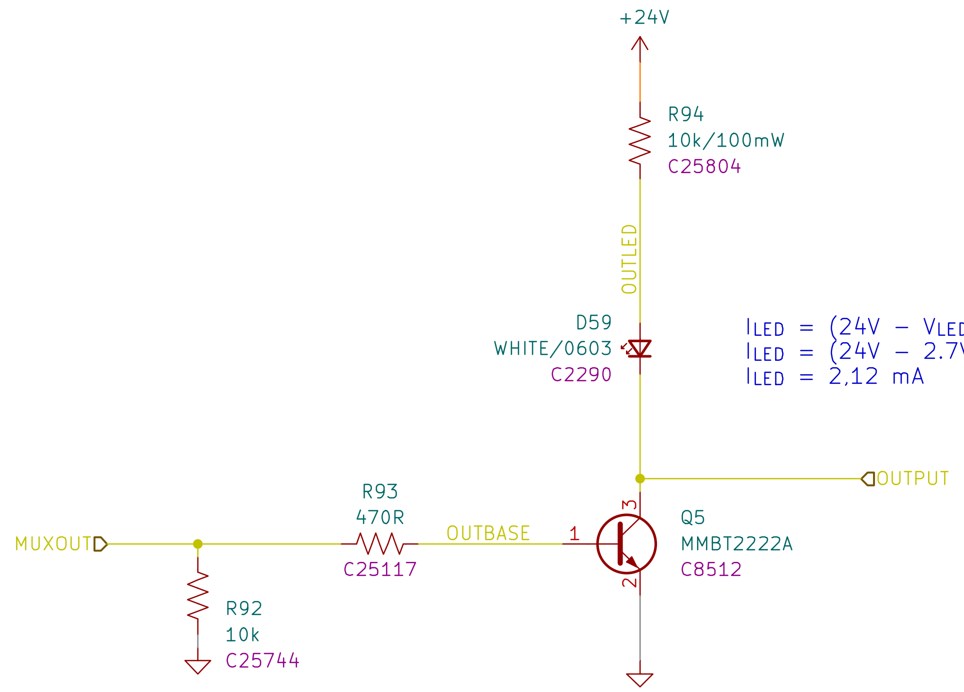
RESISTORS: 62,5mW WHERE NOT SPECIFIED



$I_{LED} = (24V - V_{LED} - V_{CE}) / 10k$
$I_{LED} = (24V - 2.7V - 0{,}1V) / 10k$
$I_{LED} = 2{,}12$ mA

$I_{BASE} = (3.3V - V_{BE}) / 470R$
$I_{BASE} = (3.3V - 0.7V) / 470R$
$I_{BASE} = 5{,}53$ mA
ADEQUATE FOR FULL SATURATION OF THE
TRANSISTOR UP TO I_C AROUND 550 mA
CONSIDERING A FORCED BETA OF 10.

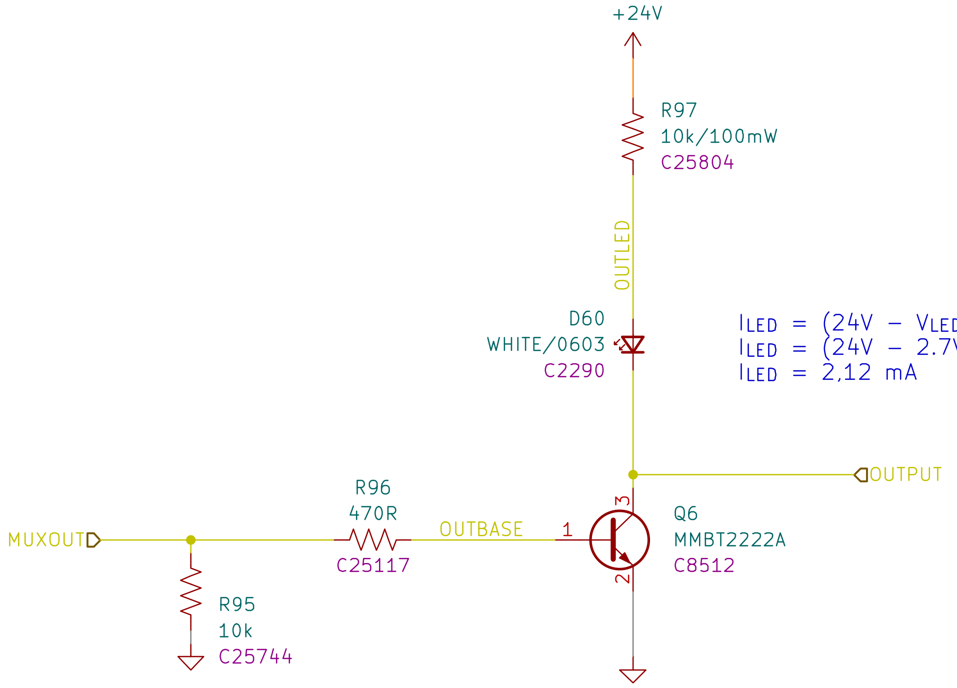
RESISTORS: 62,5mW WHERE NOT SPECIFIED



$I_{LED} = (24V - V_{LED} - V_{CE}) / 10k$
$I_{LED} = (24V - 2.7V - 0{,}1V) / 10k$
$I_{LED} = 2{,}12$ mA

$I_{BASE} = (3.3V - V_{BE}) / 470R$
$I_{BASE} = (3.3V - 0.7V) / 470R$
$I_{BASE} = 5{,}53$ mA
ADEQUATE FOR FULL SATURATION OF THE TRANSISTOR UP TO I_C AROUND 550 mA CONSIDERING A FORCED BETA OF 10.

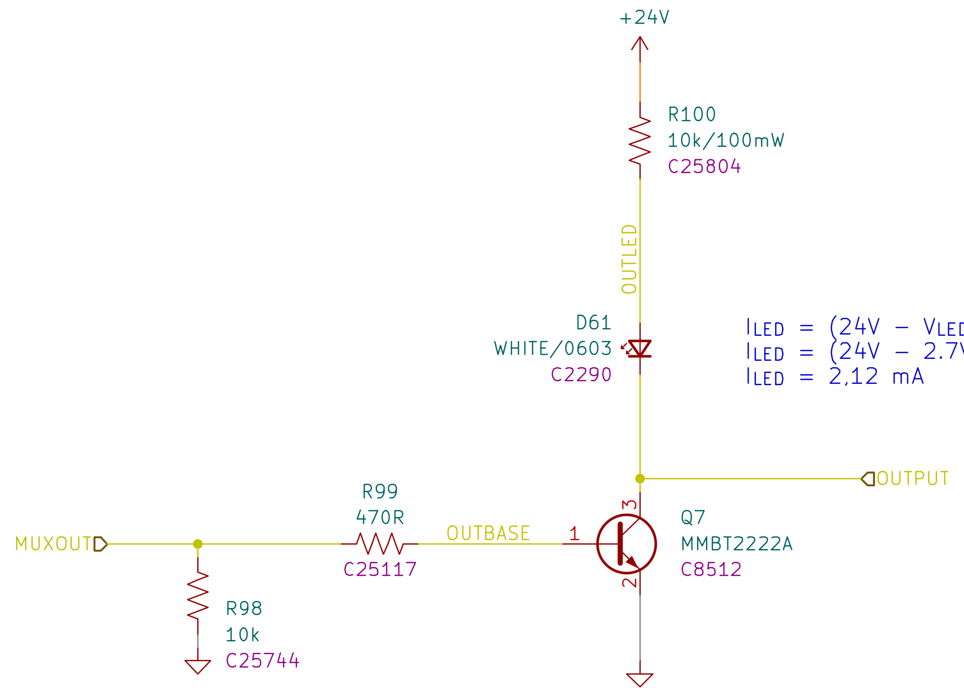
RESISTORS: 62,5mW WHERE NOT SPECIFIED



$I_{LED} = (24V - V_{LED} - V_{CE}) / 10k$
$I_{LED} = (24V - 2.7V - 0{,}1V) / 10k$
$I_{LED} = 2{,}12\ mA$

$I_{BASE} = (3.3V - V_{BE}) / 470R$
$I_{BASE} = (3.3V - 0.7V) / 470R$
$I_{BASE} = 5{,}53\ mA$
ADEQUATE FOR FULL SATURATION OF THE TRANSISTOR UP TO I_C AROUND 550 mA CONSIDERING A FORCED BETA OF 10.

RESISTORS: 62,5mW WHERE NOT SPECIFIED



$I_{LED} = (24V - V_{LED} - V_{CE}) / 10k$
$I_{LED} = (24V - 2.7V - 0{,}1V) / 10k$
$I_{LED} = 2{,}12$ mA

$I_{BASE} = (3.3V - V_{BE}) / 470R$
$I_{BASE} = (3.3V - 0.7V) / 470R$
$I_{BASE} = 5{,}53$ mA
ADEQUATE FOR FULL SATURATION OF THE
TRANSISTOR UP TO I_C AROUND 550 mA
CONSIDERING A FORCED BETA OF 10.

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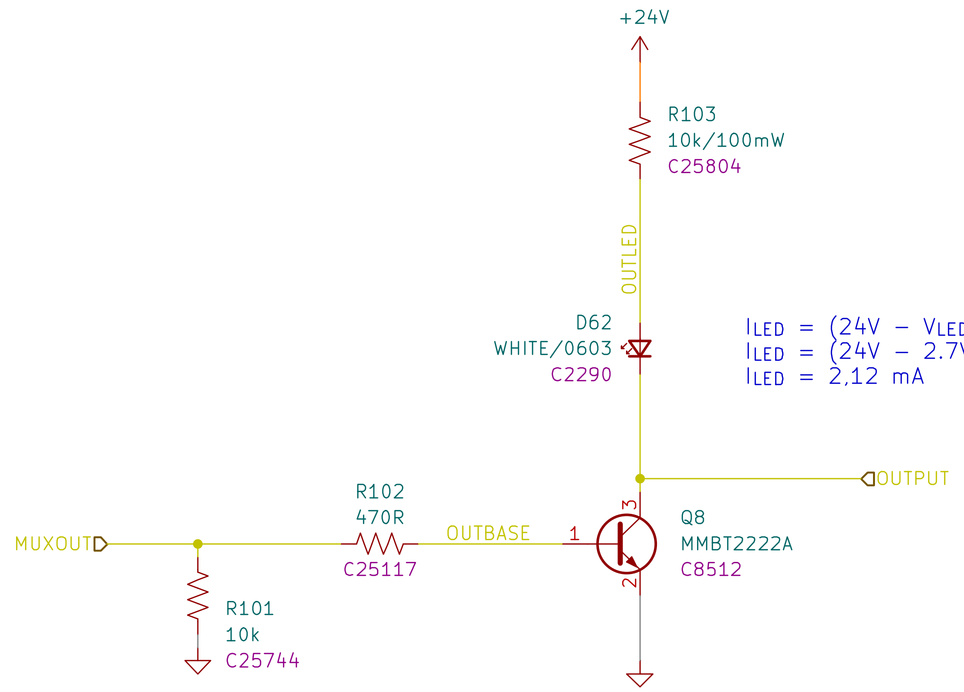

$I_{LED} = (24V - V_{LED} - V_{CE}) / 10k$
$I_{LED} = (24V - 2.7V - 0,1V) / 10k$
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ADEQUATE FOR FULL SATURATION OF THE TRANSISTOR UP TO I_C AROUND 550 mA CONSIDERING A FORCED BETA OF 10.

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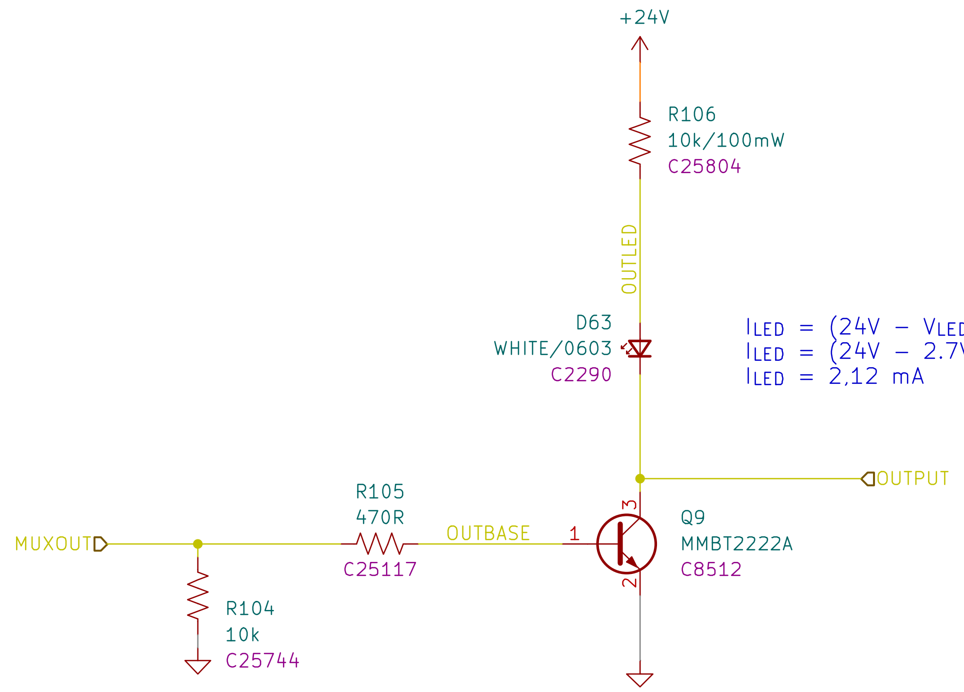

I_{LED} = (24V − V_{LED} − V_{CE}) / 10k
I_{LED} = (24V − 2.7V − 0,1V) / 10k
I_{LED} = 2,12 mA

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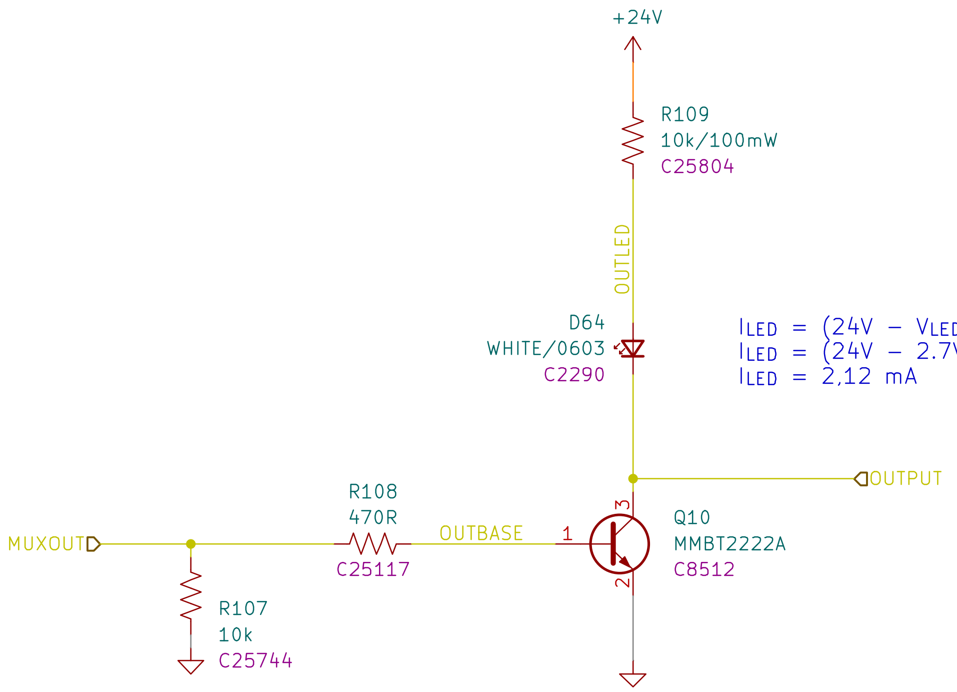

I_{LED} = (24V − V_{LED} − V_{CE}) / 10k
I_{LED} = (24V − 2.7V − 0,1V) / 10k
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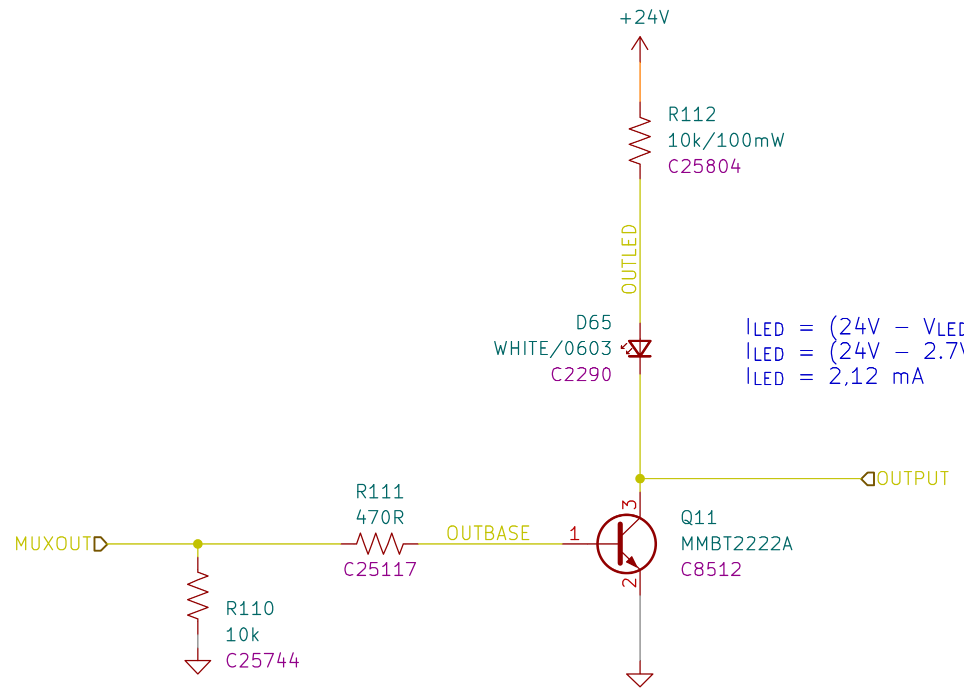

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$I_{LED} = (24V - 2.7V - 0{,}1V) / 10k$
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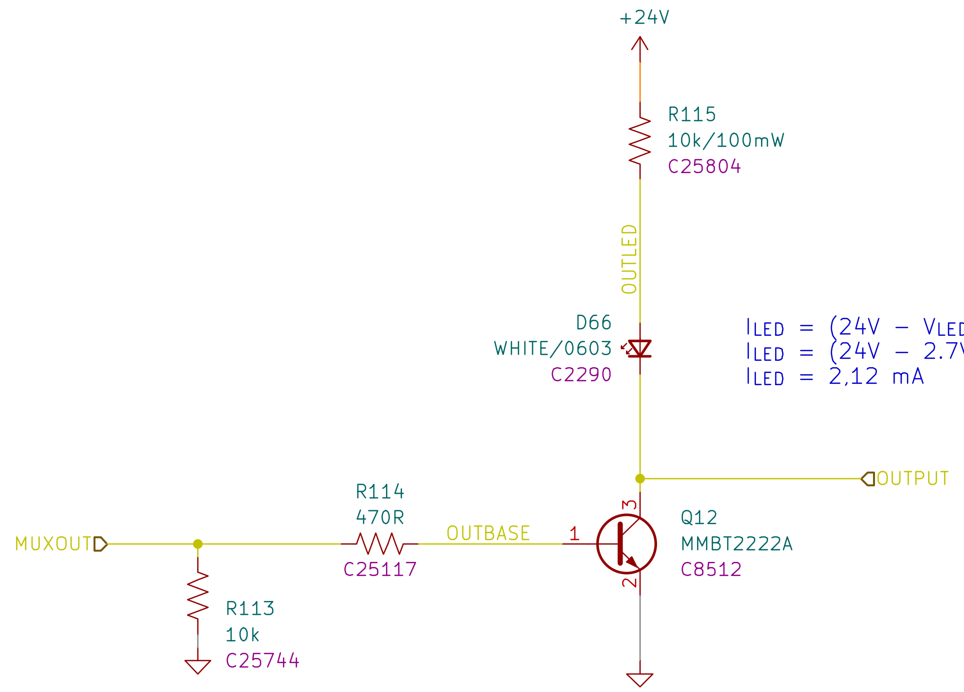

I_{LED} = (24V − V_{LED} − V_{CE}) / 10k
I_{LED} = (24V − 2.7V − 0,1V) / 10k
I_{LED} = 2,12 mA

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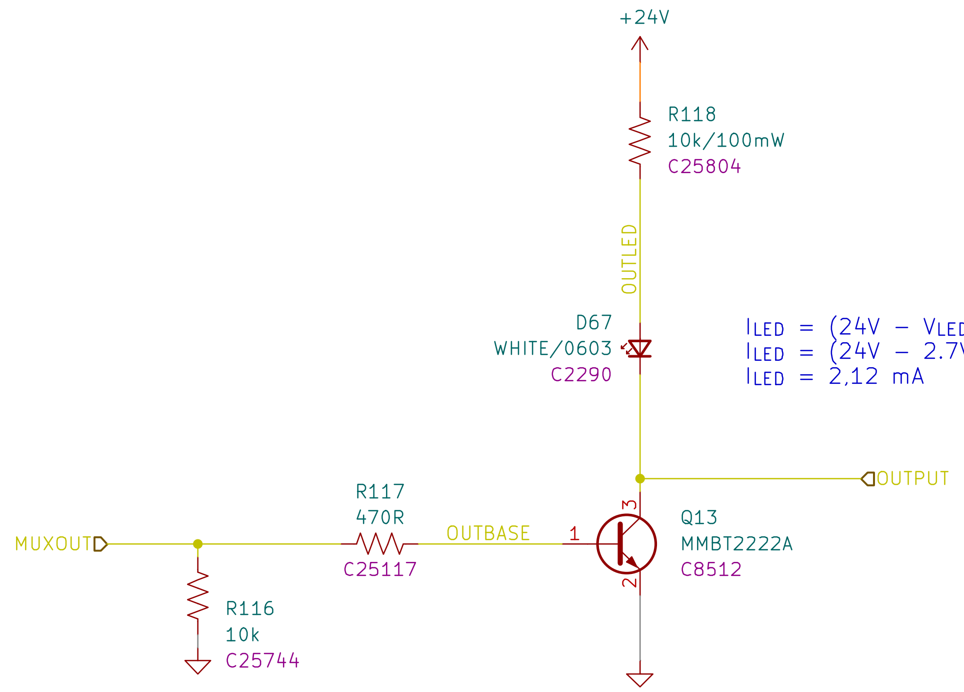

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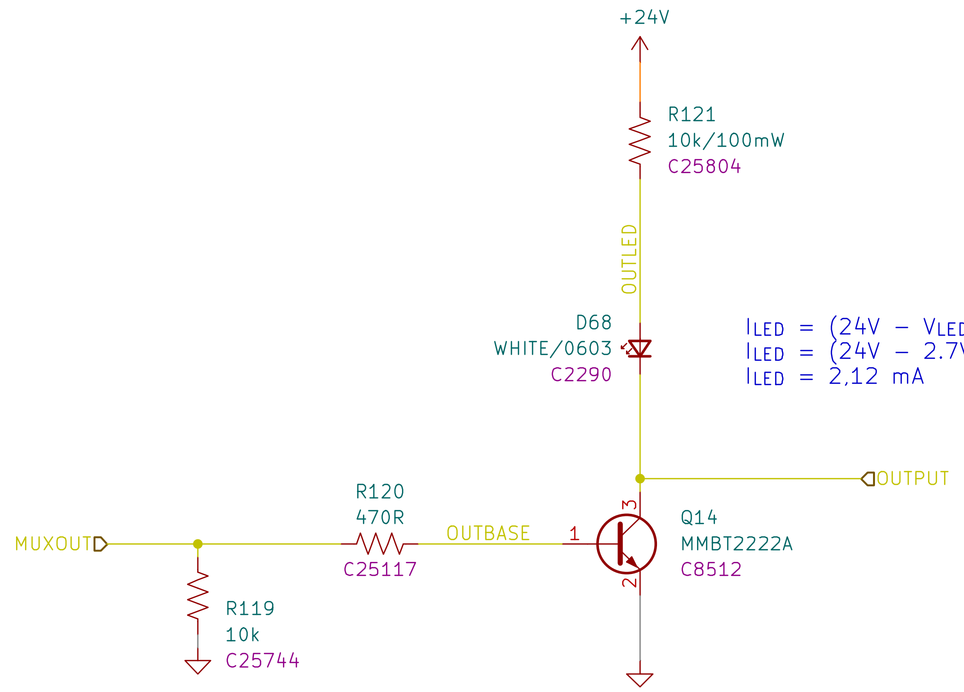

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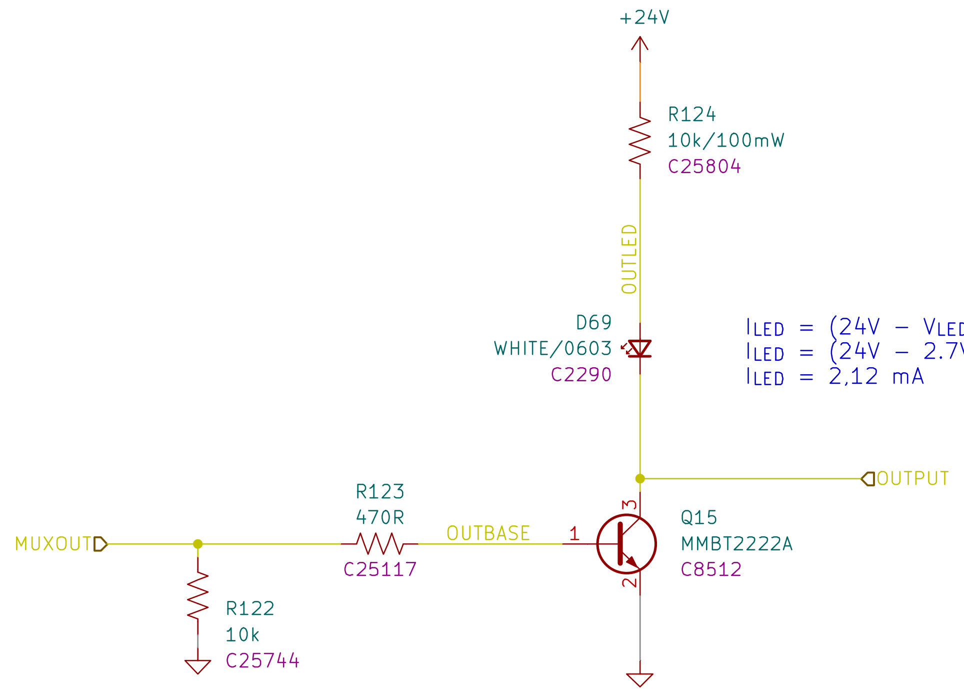

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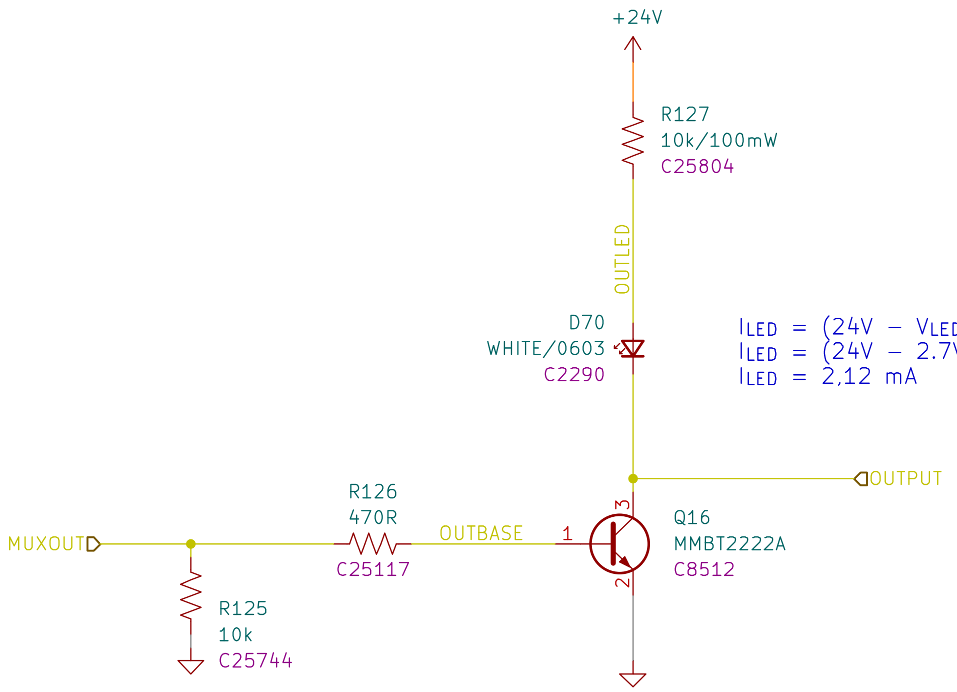

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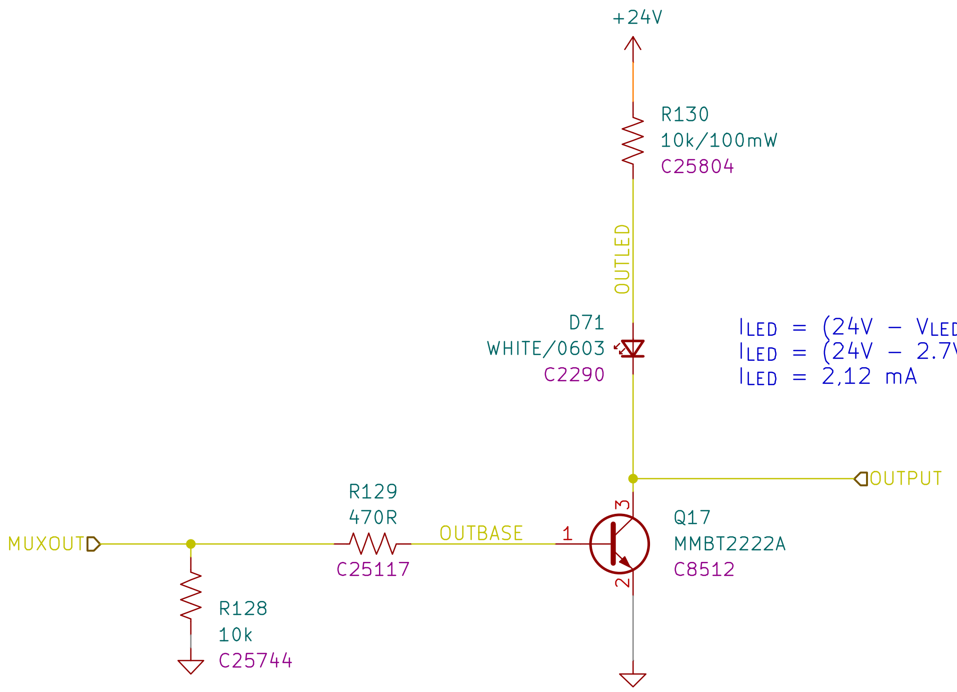

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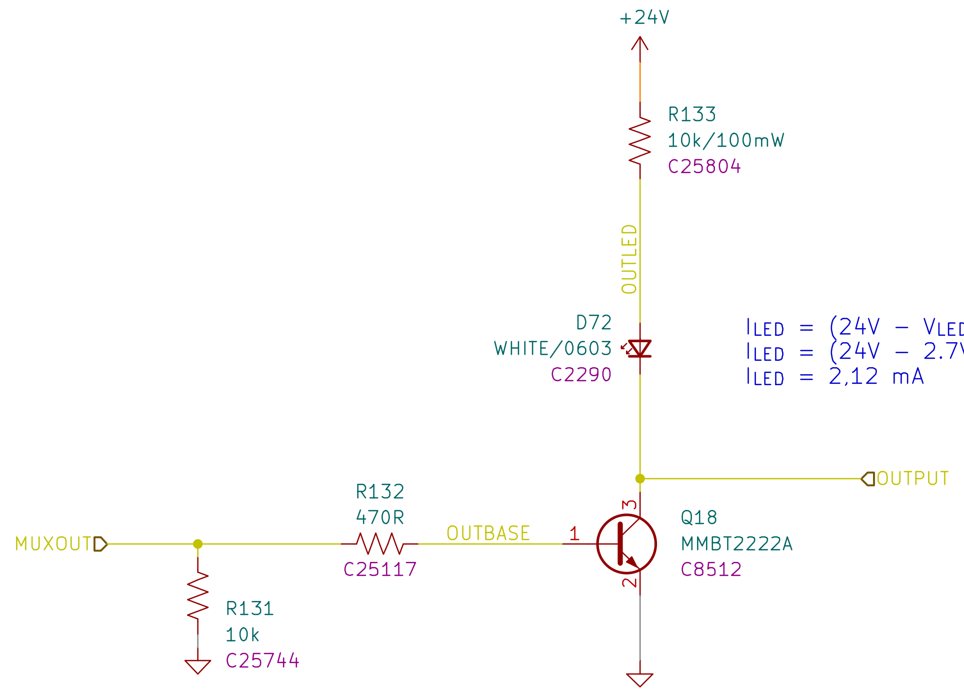

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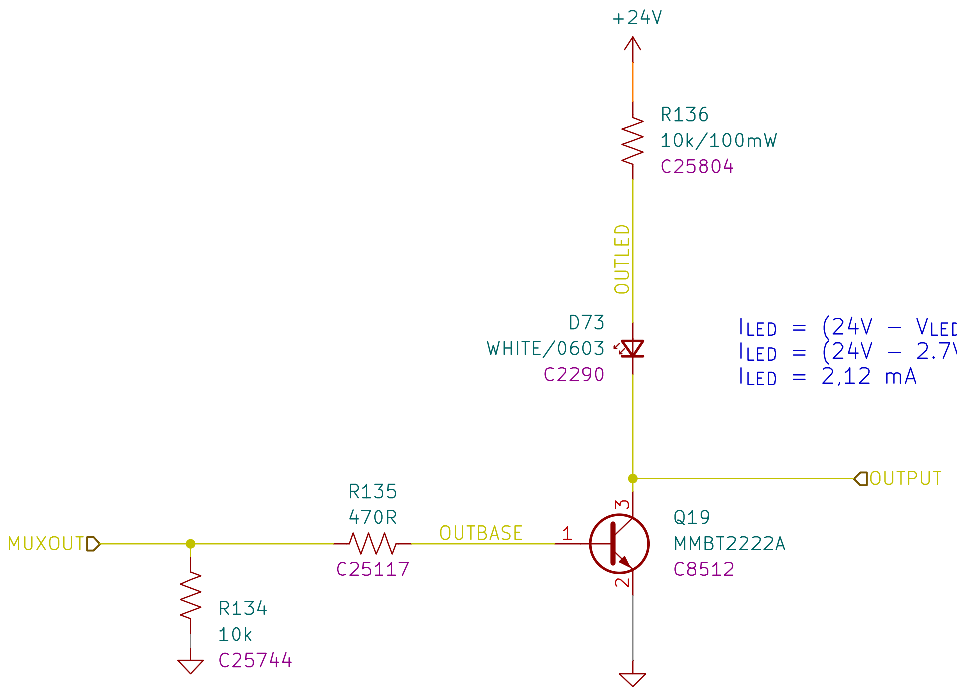

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